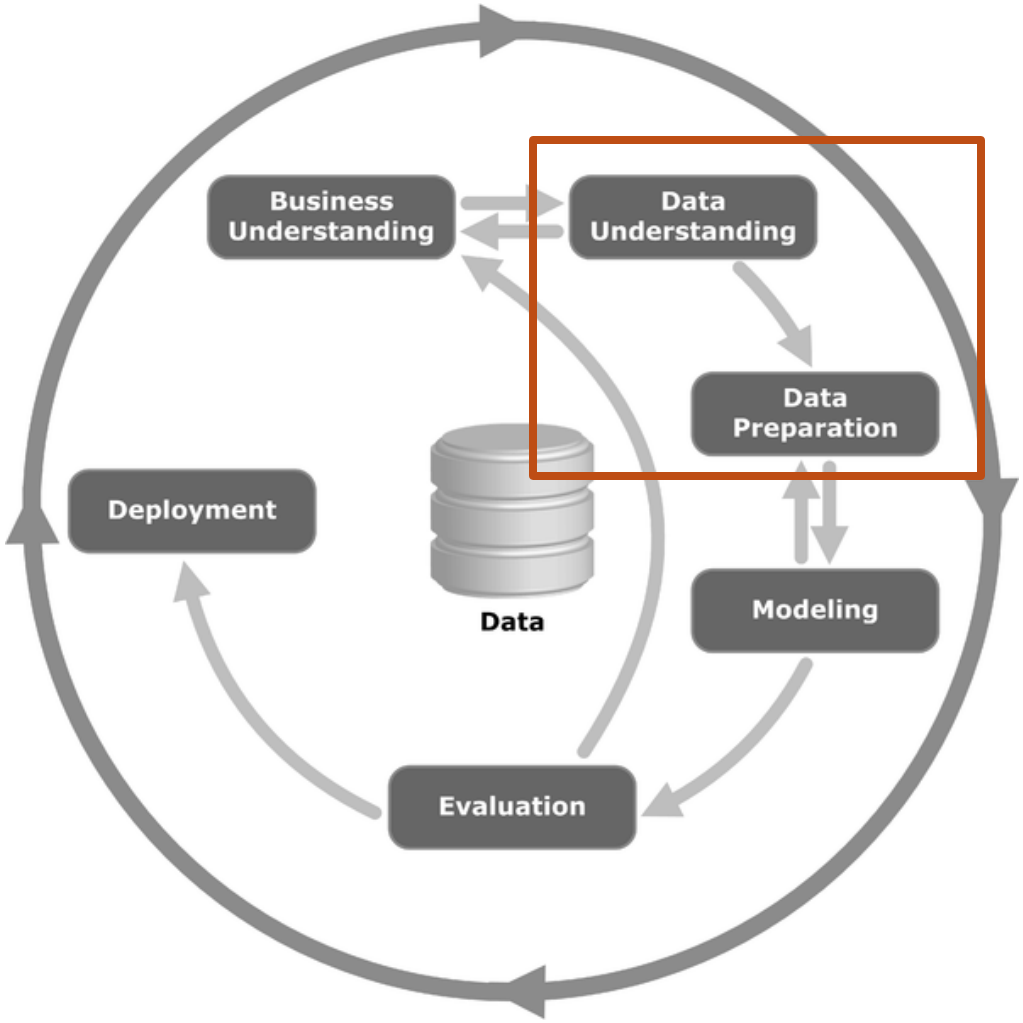
Deep Learning

Data



Tasks in the CRISP-DM Reference Model



Topics

- **Attributes/Features**
- Types of Data Sets
- Data Quality
- Data Preprocessing
- Similarity and Dissimilarity
- Density



What is Data?

- Collection of data objects and their attributes
- An attribute (in Data Mining and Machine learning often "feature") is a property or characteristic of an object
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as variable, field, characteristic
- A collection of attributes describe an object
  - Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Types of Attributes - Scales

- There are different types of attributes
  - Nominal
    - Examples: ID numbers, eye color, zip codes
  - Ordinal
    - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
  - Interval
    - Examples: calendar dates, temperatures in Celsius or Fahrenheit.
  - Ratio
    - Examples: temperature in Kelvin, length, time, counts

Categorical, Qualitative

Quantitative

Attribute Type	Description	Examples	Operations
Nominal	The values of a nominal attribute are just different **names or labels**, i.e., nominal attributes provide only enough information to distinguish one object from another.	zip codes, employee ID numbers, eye color, sex: {male, female}	$=, \neq$ mode, entropy, contingency correlation, χ^2 test
Ordinal	The values of an ordinal attribute provide enough information to **order objects**.	zip codes, employee ID numbers, eye color, sex: {male, female}	Nominal + $<, >$ median, percentiles, rank correlation, run tests, sign tests
Interval	For interval attributes, the **differences between values** are meaningful, i.e., a unit of measurement exists.	calendar dates, temperature in Celsius or Fahrenheit	Ordinal + $+, -$ mean, standard deviation, Pearson's correlation, t and F tests
Ratio	For ratio variables, both differences and **ratios are meaningful**. Double the number means twice as much.	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current	Interval + $*, /$ geometric mean, harmonic mean, percent variation

Discrete and Continuous Attributes

- **Discrete Attribute**
  - Has only a finite or countably infinite set of values.
  - Binary attributes are a special case of discrete attributes.
  - **Examples**: zip codes, counts, or the set of words in a collection of documents
  - **Representation**: Strings or integer variables (enumeration type).
- **Continuous Attribute**
  - Has real numbers as attribute values
  - **Examples**: temperature, height, or weight.
  - **Representation**: floating-point variables. Computers represent real numbers in a discrete format with a finite number of digits.

Examples

- What is the scale of measurement of:
  - —Number of cars per minute (count data)
  - —Age data grouped in:
    0-4 years, 5-9, 10-14, …
  - —Age data grouped in:
    <20 years, 21-30, 31-40, 41+

Topics

- Attributes/Features
- **Types of Data Sets**
- Data Quality
- Data Preprocessing
- Similarity and Dissimilarity
- Density



Types of data sets

- Record
  - —Data Matrix
  - —Document Data
  - —Transaction Data
- Graph
  - —World Wide Web
  - —Molecular Structures
- Ordered
  - —Spatial Data
  - —Temporal Data
  - —Sequential Data
  - —Genetic Sequence Data

Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes (e.g., from a relational database)

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

n attributes

m objects

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
5.6	2.7	4.2	1.3
6.5	3.0	5.8	2.2
6.8	2.8	4.8	1.4
5.7	3.8	1.7	0.3
5.5	2.5	4.0	1.3
4.8	3.0	1.4	0.1
5.2	4.1	1.5	0.1

Document Data

- Each document becomes a `term' vector,
  - —each term is a component (attribute) of the vector,
  - —the value of each component is the number of times the corresponding term occurs in the document.

Terms

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

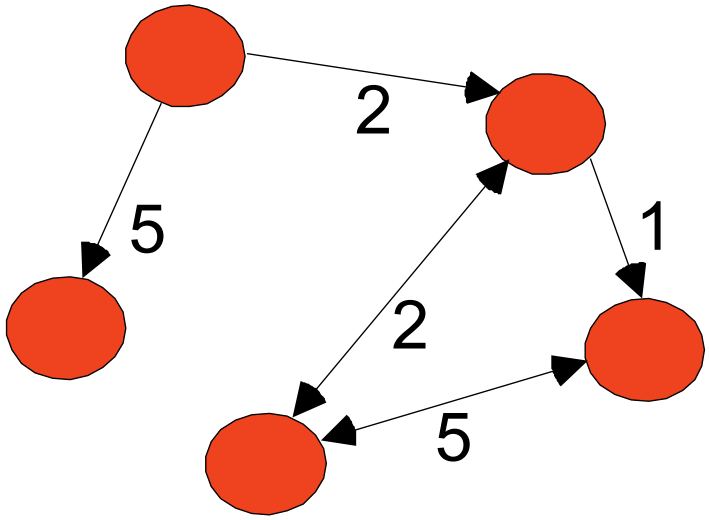
Transaction Data

- A special type of record data, where
  - —each record (transaction) involves a set of items.
  - —For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

TID	Items
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Graph Data

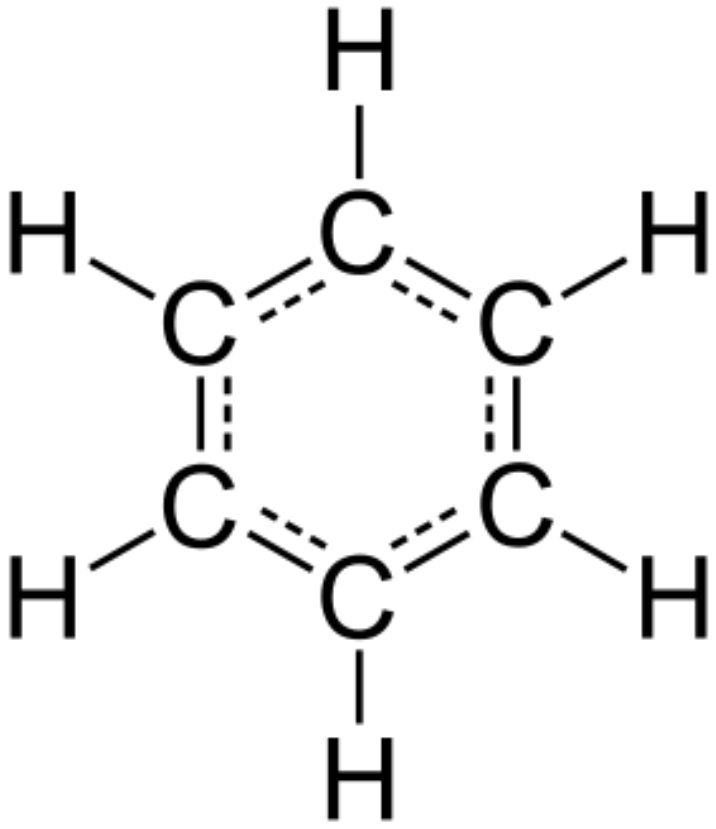
- Examples: Generic graph and HTML Links




```
<a href="papers/papers.html#bbbb">
Data Mining </a>
<li>
<a href="papers/papers.html#aaaa">
Graph Partitioning </a>
<li>
<a href="papers/papers.html#aaaa">
Parallel Solution of Sparse Linear System of Equations </a>
<li>
<a href="papers/papers.html#ffff">
N-Body Computation and Dense Linear System Solvers
```

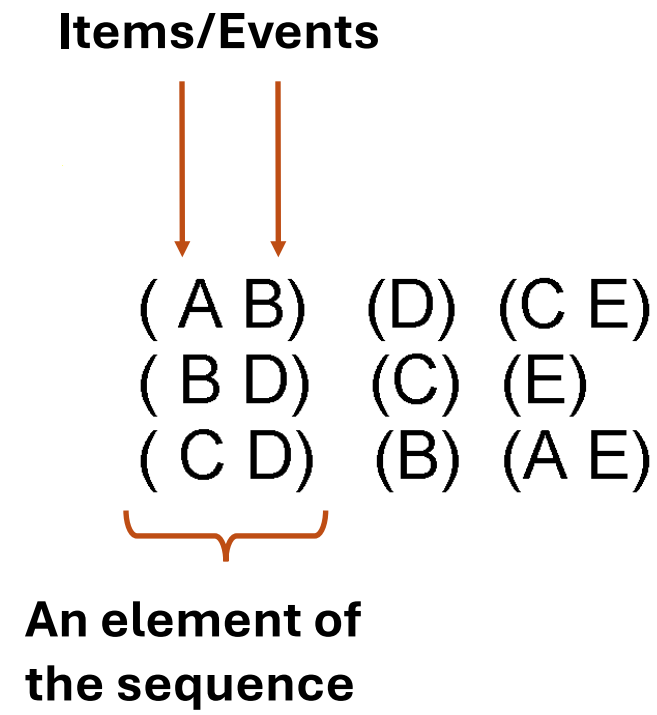
Chemical Data

- Benzene Molecule: C6H6



Ordered Data

- Sequences of transactions



Ordered Data

- Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

Subsequences

Ordered Data: Time Series Data

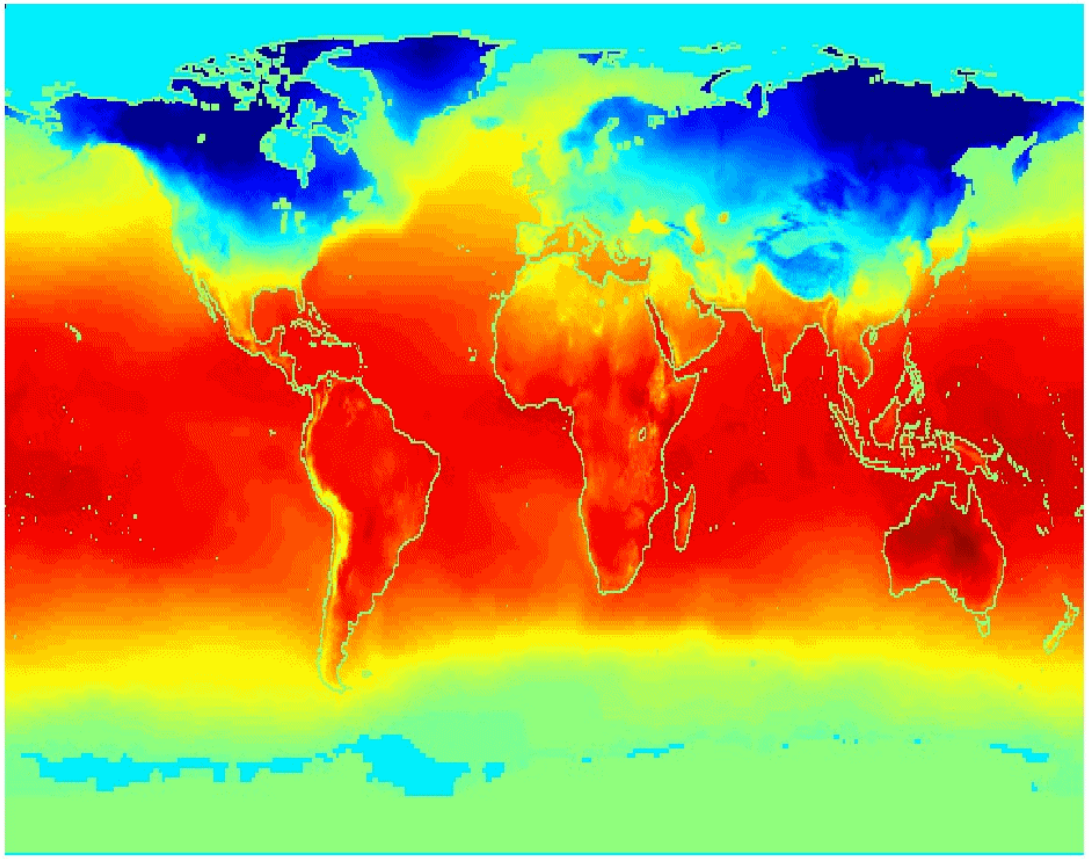


Source: FactSet

Ordered Data: Spatio-Temporal

Average Monthly Temperature of land and ocean

Jan, Feb, Mar, …



Topics

- Attributes/Features
- Types of Data Sets
- **Data Quality**
- Data Preprocessing
- Similarity and Dissimilarity
- Density



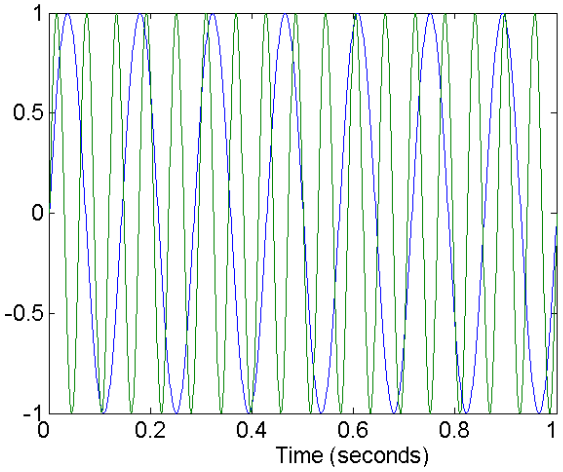
Data Quality

- What kinds of data quality problems exist?
  - Noise and outliers
  - Missing values
  - Duplicate data
- How can we detect problems with the data?
  - Statistics
  - Visualization
- What can we do about these problems?
  - Mark value as missing
  - Remove object

Noise

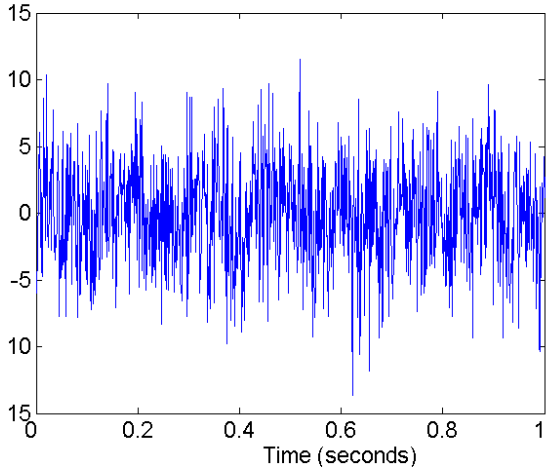
- Noise refers to modification of original values
  - —Examples: distortion of a person's voice when talking on a poor phone, "snow" on television screen, measurement errors.

Signal: Two Sine Waves



Noisy measurement

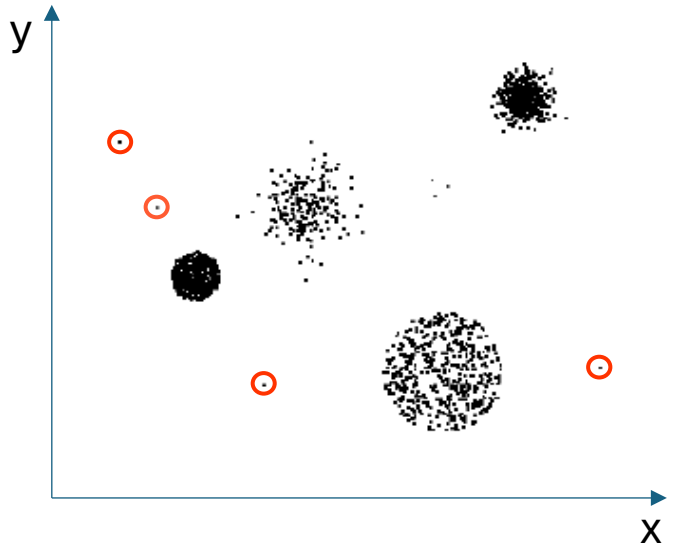
Two Sine Waves + Noise



- Find less noisy data
- Sometimes we can de-noise (signal processing)

Outliers

- Outliers are data objects with feature values that are considerably different than most of the other data objects in the data set.
- Reasons:
  - A true outlier is a special object (e.g., a genius's IQ score).
  - May be the result of a measurement mistake.



- Typical treatment: Statistical outlier detection +
  - Make outlying feature missing, or
  - Remove the complete outlier object

Missing Values

- Reasons for missing values
  - —Information is not collected (e.g., people decline to give their age and weight)
  - —Attributes may not be applicable to all cases (e.g., annual income is not applicable to children)
  - —Value was a mistake and set to missing.
- Handling missing values
  - —Eliminate data objects with missing value.
  - —Eliminate feature with missing values.
  - —Ignore the missing value during analysis.
  - —Estimate missing values = Imputation (e.g., replace with mean or weighted mean where all possible values are weighted by their probabilities)

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	NA	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Duplicate Data

- Data set may include data objects that are duplicates, or "close duplicates" of one another
  - —Major issue when merging data from heterogeneous sources
- Examples:
  - —Same person with multiple email addresses
- Data cleaning
  - —Process of dealing with duplicate data issues
  - —ETL tools typically support deduplication

Topics

- Attributes/Features
- Types of Data Sets
- Data Quality
- **Data Preprocessing**
- Similarity and Dissimilarity
- Density



Data Preprocessing

- Aggregation
- Sampling
- Dimensionality Reduction
- Feature subset selection
- Feature creation
- Discretization and Binarization
- Attribute Transformation

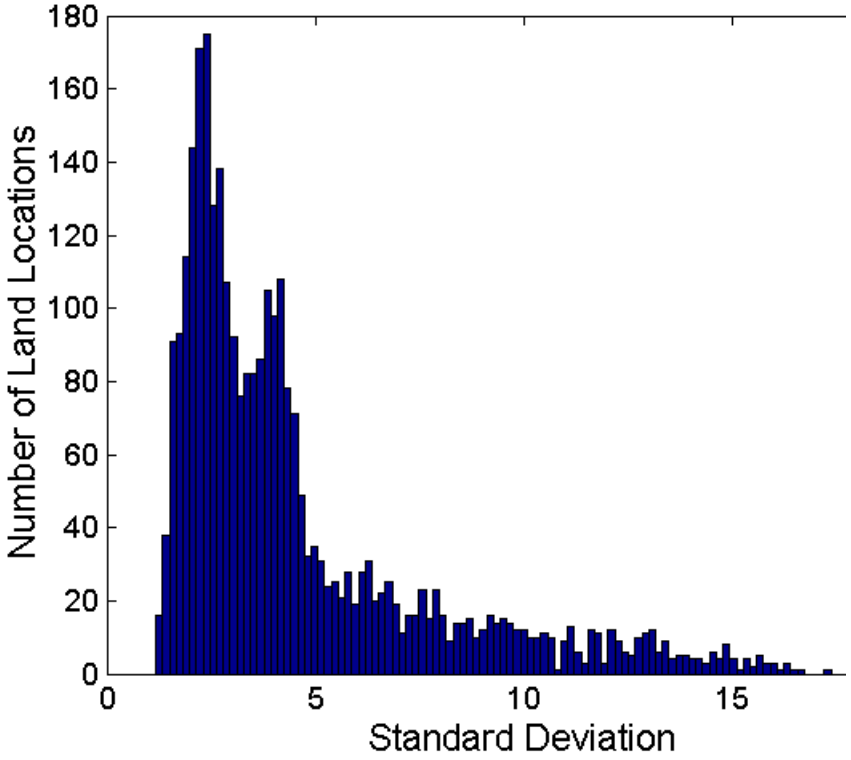
Aggregation

- Combining two or more attributes (or objects) into a single attribute (or object)

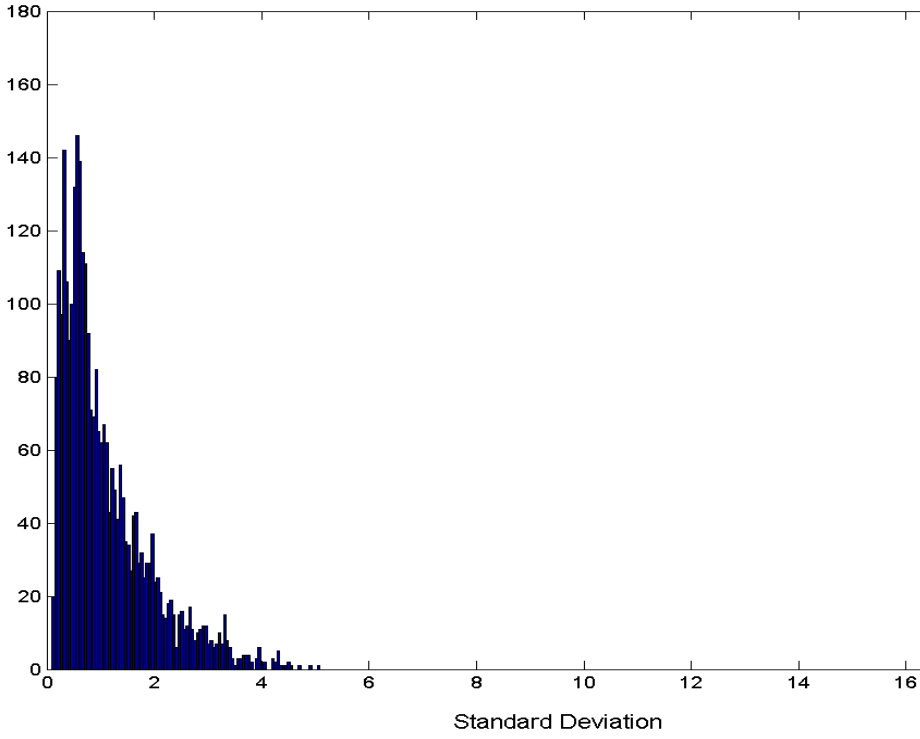
- Purpose
  - —Data reduction
    - Reduce the number of attributes or objects
  - —Change of scale
    - Cities aggregated into regions, states, countries, etc
  - —More “stable” data
    - Aggregated data tends to have less variability (e.g., reduce seasonality by aggregation to yearly data)

Aggregation

Variation of Precipitation in Australia



Standard Deviation of Average Monthly Precipitation



Standard Deviation of Average Yearly Precipitation

Sampling

- Sampling is the main technique employed for data selection.
  - It is often used for both the preliminary investigation of the data and the final data analysis.

- Statisticians sample because **obtaining** the entire set of data of interest is too expensive or time consuming.

- Sampling is used in data mining because **processing** the entire set of data of interest is too expensive (e.g., does not fit into memory or is too slow).

Sampling …

- The key principle for effective sampling is the following:
  - —using a sample will work almost as well as using the entire data sets, if the sample is **representative**.
  - —A sample is representative if it has approximately the same property (of interest) as the original set of data.

Types of Sampling

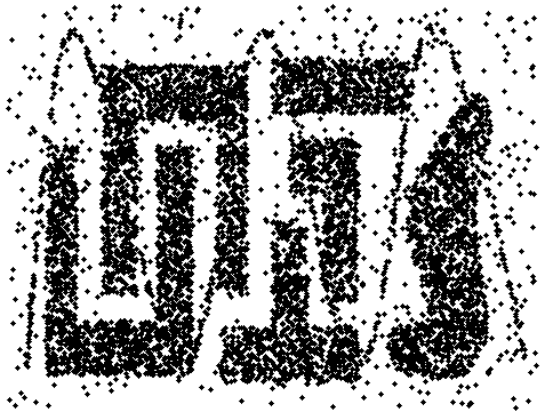
Replacement?

- **Sampling without replacement**
  As each item is selected, it is removed from the population.
- **Sampling with replacement**
  Objects are not removed from the population as they are selected for the sample. Note: the same object can be picked up more than once.

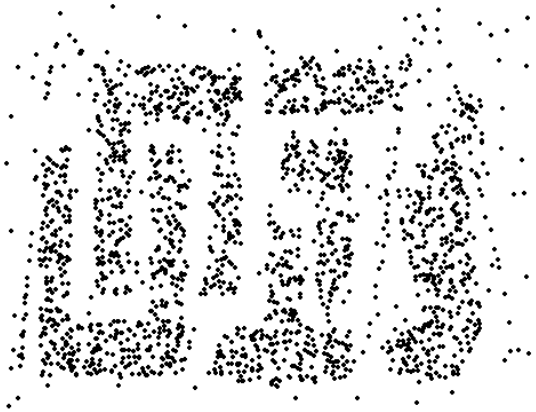
Selection?

- **Simple random sampling**
  There is an equal probability of selecting any particular item.
- **Stratified sampling**
  Split the data into several partitions; then draw random samples from each partition.

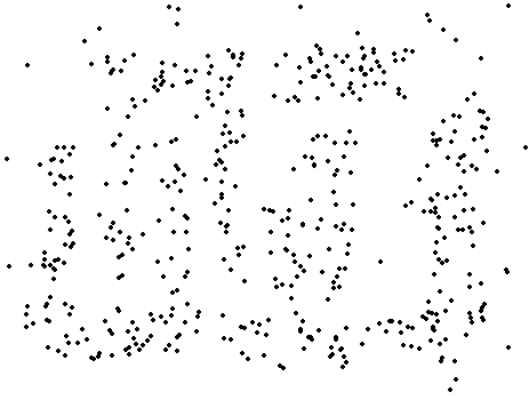
Sample Size



8000 points



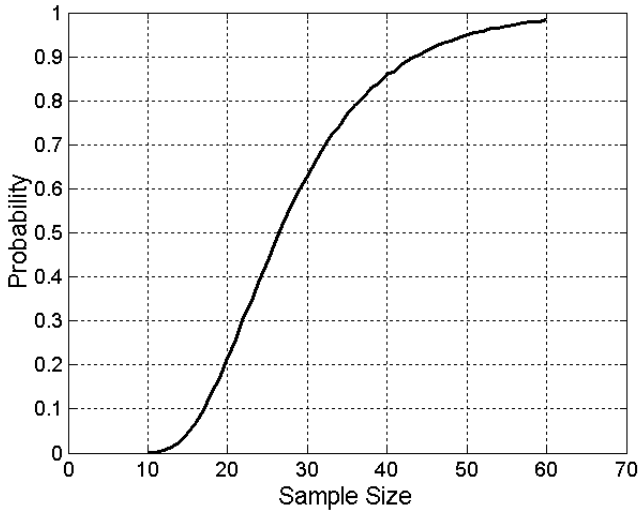
2000 Points



500 Points

Sample Size

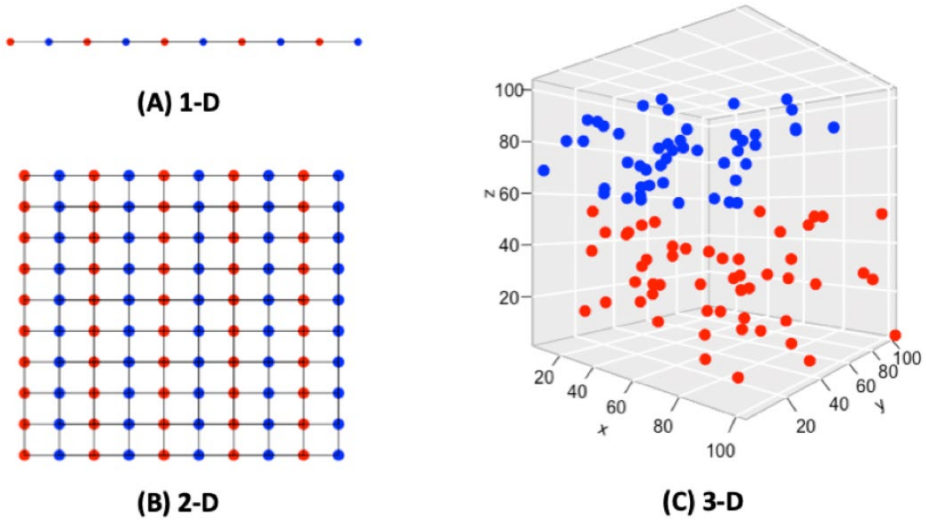
- What sample size is necessary to get at least one object from each of 10 groups.



- Sample size determination:
  - —Statistics: confidence interval for parameter estimate or desired statistical power of test.
  - —Machine learning: often more is better, cross-validated accuracy.

Curse of Dimensionality

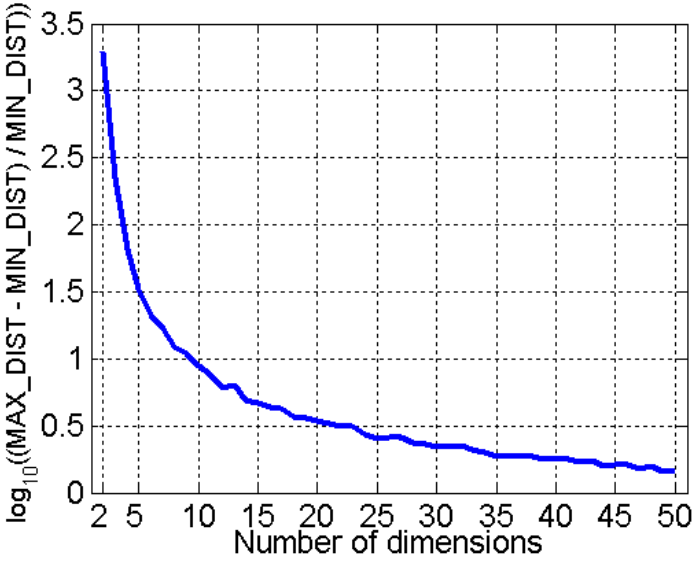
- When dimensionality increases, the size of the data space grows exponentially.



Points and space

- Definitions of density and distance between points, which is critical for clustering and outlier detection, become less meaningful
  - Density → 0
  - All points tend to have the same Euclidean distance to each other.

Experiment: Randomly generate 500 points. Compute difference between max and min distance between any pair of points



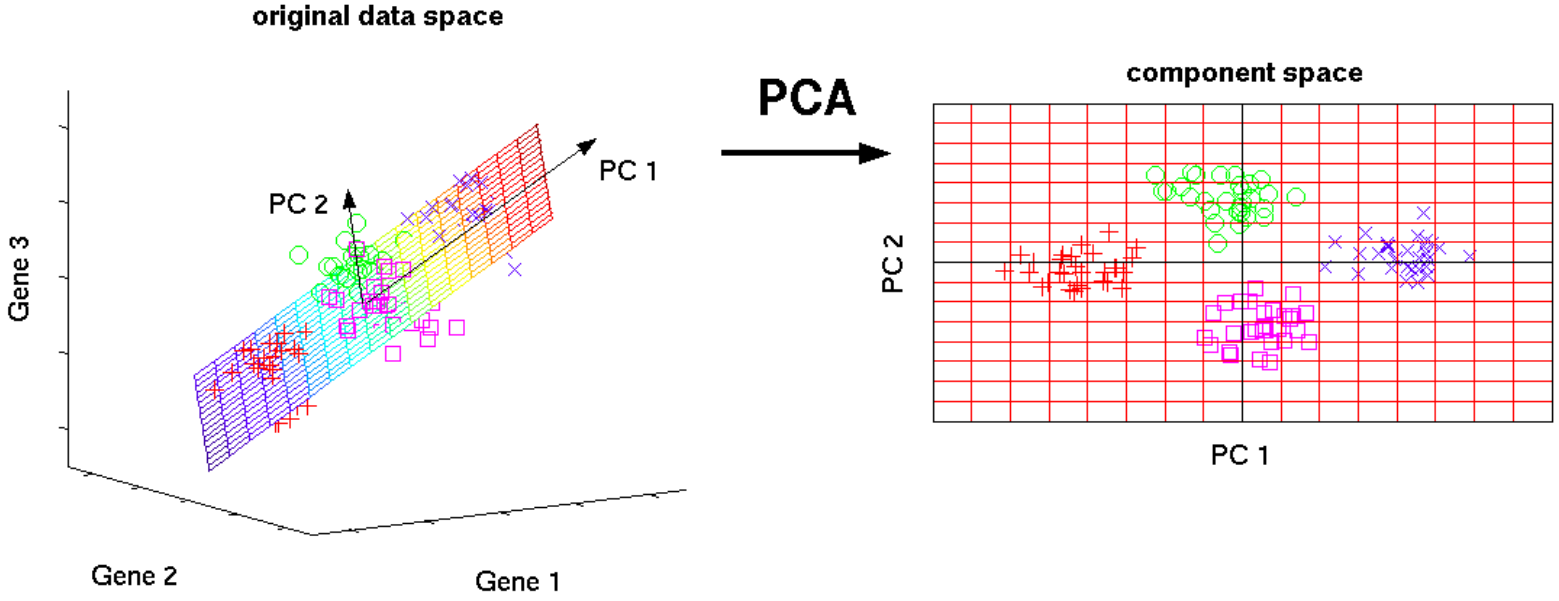
Dimensionality Reduction

- Purpose:
  - —Avoid curse of dimensionality
  - —Reduce amount of time and memory required by data mining algorithms
  - —Allow data to be more easily visualized
  - —May help to eliminate irrelevant features or reduce noise

- Techniques
  - —Principle Component Analysis
  - —Singular Value Decomposition
  - —Others: supervised and non-linear techniques

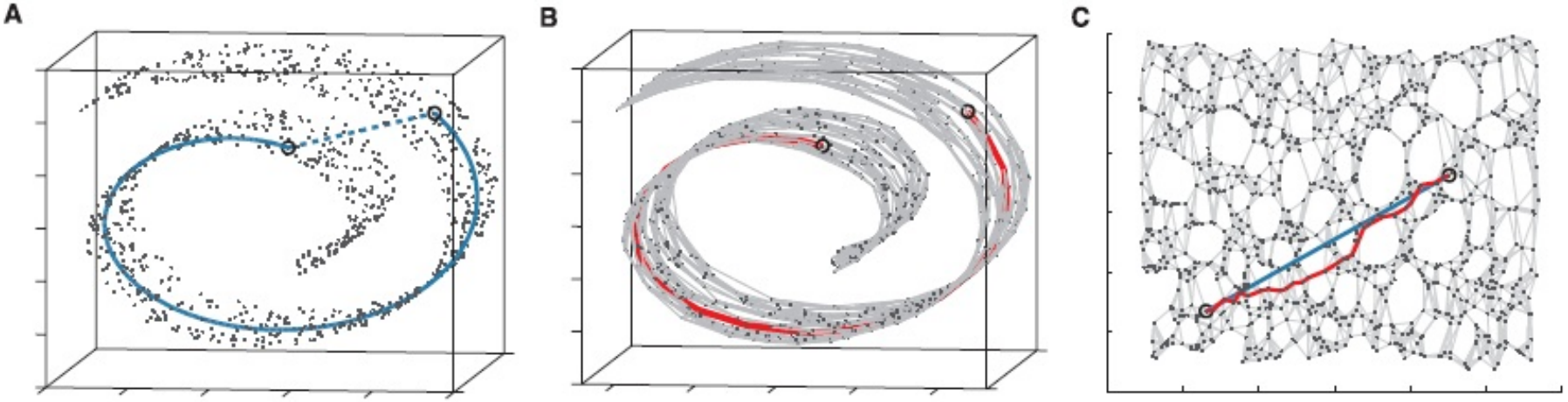
Dimensionality Reduction: Principal Components Analysis (PCA)

- **Goal**: Map points to a lower dimensional space while preserving distance information.



- **Method**: Find a projection (new axes) that captures the largest amount of variation in data. This can be done using eigenvectors of the covariance matrix or SVD (singular value decomposition).

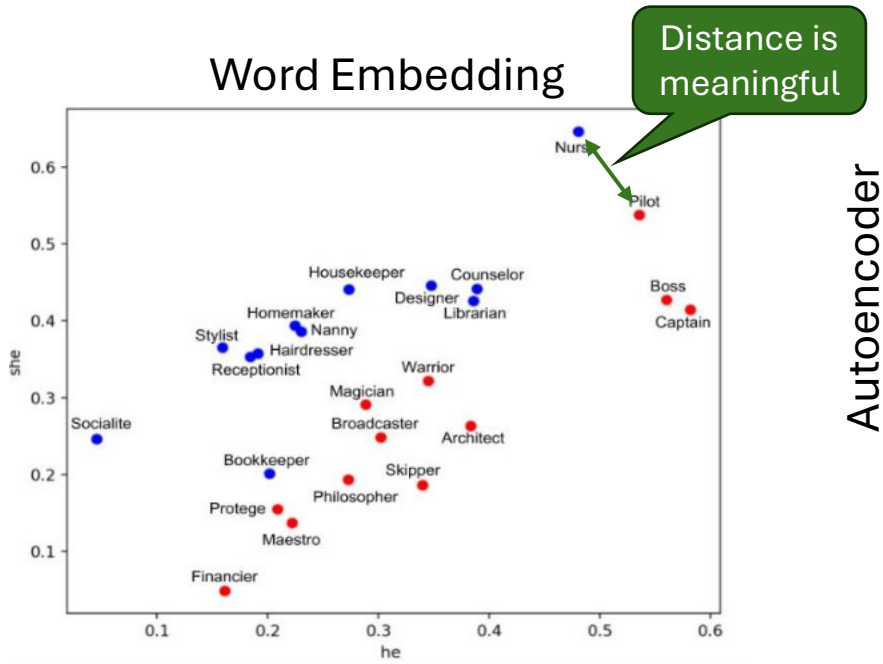
Dimensionality Reduction: ISOMAP

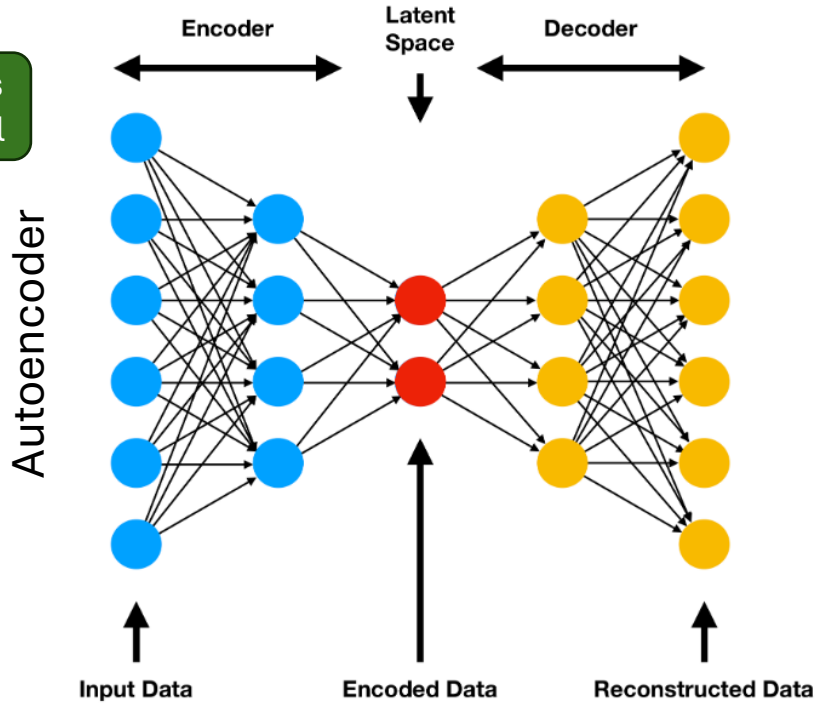


- **Goal**: Unroll the “swiss roll!“ (i.e., preserve distances on the roll)
- **Method**: Use a non-metric space, i.e., distances are not measured by Euclidean distance, but along the surface of the roll (geodesic distances).
  1. Construct a neighbourhood graph (k-nearest neighbors or within a radius).
  2. For each pair of points in the graph, compute the shortest path distances = geodesic distances.
  3. Create a lower dimensional embedding using the geodesic distances (multi-dimensional scaling; MDS)

Low-dimensional Embedding

- General notion of representing objects described in one space (i.e., set of features) in a different space using a map $f : X \rightarrow Y$
- PCA is an example where Y is the space spanned by the principal components and objects close in the original space X are embedded in space Y.
- Low-dimensional embeddings can be produced with various other methods:
  - —T-SNA: T-distributed Stochastic Neighbor Embedding; non-linear for visualization of high-dimensional datasets.
  - —Autoencoders (deep learning): non-linear
  - —Word embedding: Word2vec, GloVe, BERT





Feature Subset Selection

= Remove features (columns):

- Redundant features
  - duplicate information contained in multiple features (are correlated)
  - Example: purchase price of a product and the amount of sales tax paid
- Irrelevant features
  - contain no information that is useful for the data mining task
  - Example: students' ID is often irrelevant to the task of predicting students' GPA

Methods

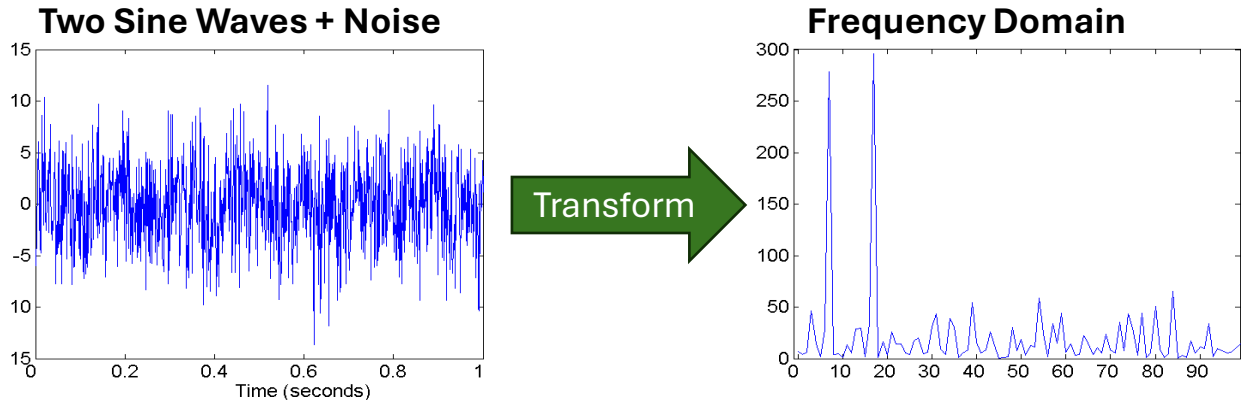
- Embedded approaches:
  - Feature selection occurs naturally as part of the data mining algorithm (e.g., regression, decision trees).
- Filter approaches:
  - Features are selected before data mining algorithm is run
  - (e.g., highly correlated features)
- Brute-force approach:
  - Try all possible feature subsets as input to data mining algorithm and choose the best.
- Wrapper approaches:
  - Use the data mining algorithm as a black box to find best subset of attributes (often using greedy search)

Feature Creation

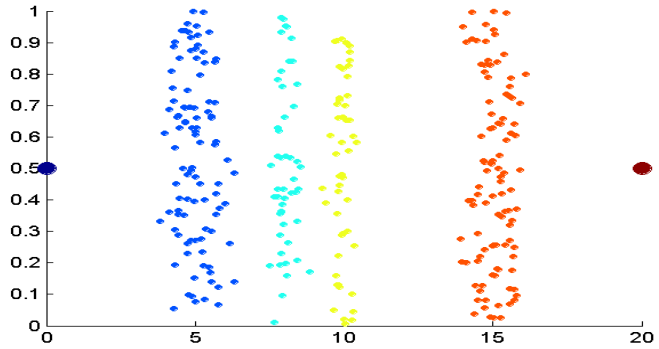
Create new attributes that can capture the important information in a data set much more efficiently than the original attributes

Three general methodologies

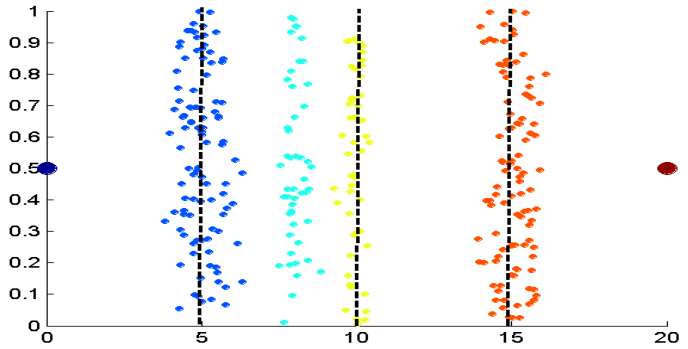
- Feature Extraction
  - Domain-specific (e.g., face recognition in image mining)
- Feature Construction / Feature Engineering
  - Combining features (interactions: multiply features)
  - Example: Calculate the body mass index from height and weight
- Mapping Data to New Space
  - Example: Fourier transform/Wavelet transform



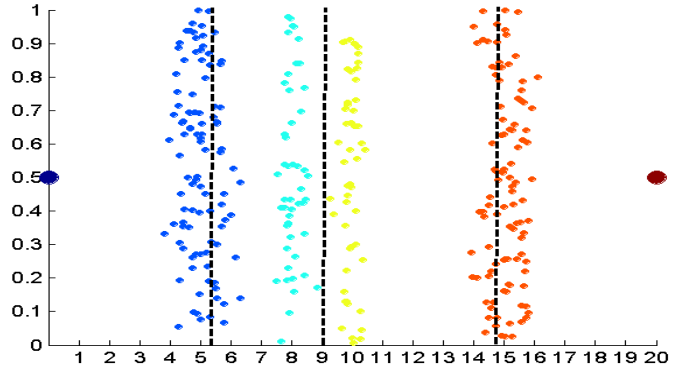
Mapping Data to New Space: Unsupervised Discretization



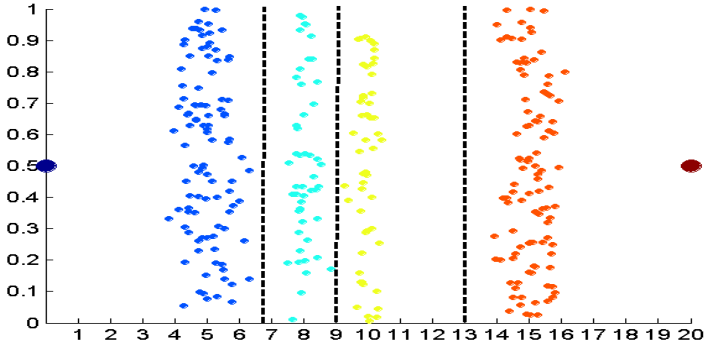
Data



Equal interval width



Equal frequency



K-means

Attribute Transformation: Normalization

- A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values
  - —Simple functions: $x^k, \log(x), e^x, |x|$
  - —Standardization and Normalization
    The z-score normalizes data roughly to an interval of $[-3,3]$.

$$x' = \frac{x - \bar{x}}{s_x}$$

$\bar{x}$ ... column (attribute) mean

s_x ... column (attribute) standard deviation

Topics

- Attributes/Features
- Types of Data Sets
- Data Quality
- Data Preprocessing
- **Similarity and Dissimilarity**
- Density



Similarity and Dissimilarity

- Similarity
  - —Numerical measure of how alike two data objects are.
  - —Is higher when objects are more alike.
  - —Often falls in the range [0,1]
- Dissimilarity
  - —Numerical measure of how different are two data objects
  - —Lower when objects are more alike
  - —Minimum dissimilarity is often 0
  - —Upper limit varies
- Proximity refers to a similarity or dissimilarity

Similarity/Dissimilarity for Simple Attributes

p and *q* are the attribute values for two data objects.

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } p = q \\ 1 & \text{if } p \neq q \end{cases}$	$s = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases}$
Ordinal	$d = \frac{\lvert p-q \rvert}{n-1}$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - \frac{\lvert p-q \rvert}{n-1}$
Interval or Ratio	$d = \lvert p - q \rvert$	$s = -d$, $s = \frac{1}{1+d}$ or $s = 1 - \frac{d - min_d}{max_d - min_d}$

$$s = f(d)$$

f can be any strictly decreasing function.

Euclidean Distance

point	x	y
p	0	2
q	2	0

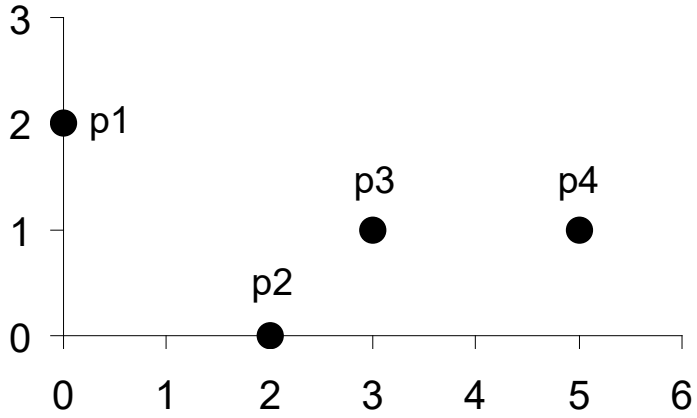
- Euclidean Distance (for quantitative attribute vectors)

$$d_E = \sqrt{\sum_{k=1}^{n} (p_k - q_k)^2} = \|\boldsymbol{p} - \boldsymbol{q}\|_2$$

  - Where $\boldsymbol{p}$ and $\boldsymbol{q}$ are two objects represented by vectors. n is the number of dimensions (attributes) of the vectors and p_k and q_k are, respectively, the kth attributes (components) or data objects p and q.
  - $\|\cdot\|_2$ is the L^2 vector norm (i.e., length of a vector in Euclidean space).

- **Note**: If ranges differ between components of $\boldsymbol{p}$ then standardization (z-scores) is necessary to avoid one variable to dominate the distance.

Euclidean Distance



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0.00	2.83	3.16	5.10
p2	2.83	0.00	1.41	3.16
p3	3.16	1.41	0.00	2.00
p4	5.10	3.16	2.00	0.00

Distance Matrix

Minkowski Distance

point	x	y
p	0	2
q	2	0

- Minkowski Distance is a generalization of Euclidean Distance

$$d_M = \left(\sum_{k=1}^{n} |p_k - q_k|^r \right)^{\frac{1}{r}} = \|\boldsymbol{p} - \boldsymbol{q}\|_r$$

  - Where $\boldsymbol{p}$ and $\boldsymbol{q}$ are two objects represented by vectors. n is the number of dimensions (attributes) of the vectors and p_k and q_k are, respectively, the kth attributes (components) or data objects p and q.

- **Note**: If ranges differ then standardization (z-scores) is necessary to avoid one variable to dominate the distance.

Minkowski Distance: Examples

- $r = 1$. City block (Manhattan, taxicab, L^1 norm) distance.
  - —A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors
- $r = 2$. Euclidean distance (L^2 norm)
- $r = \infty$. "supremum" (maximum norm, L^∞ norm) distance.
  - —This is the maximum difference between any component of the vectors
- Do not confuse r with n, i.e., all these distances are defined for all numbers of dimensions.

Minkowski Distances

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

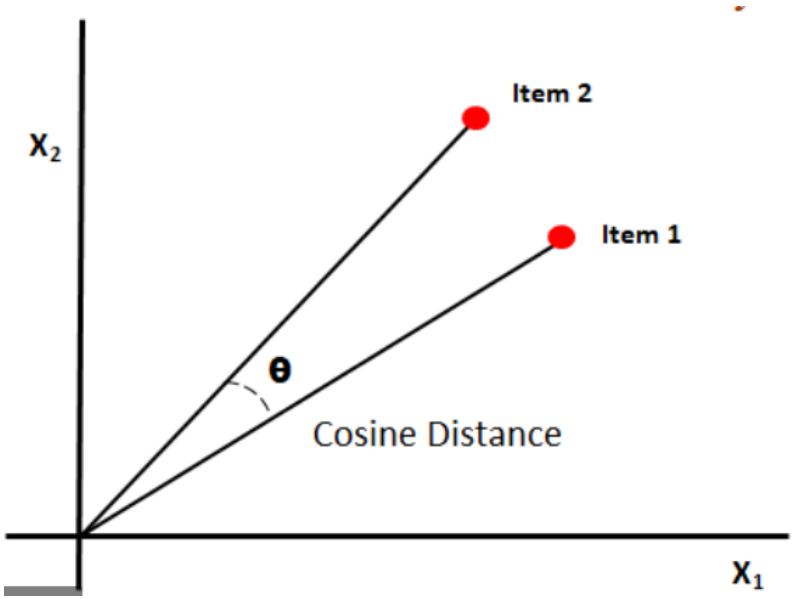
Distance Matrix

L^1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L^2	p1	p2	p3	p4
p1	0.00	2.83	3.16	5.10
p2	2.83	0.00	1.41	3.16
p3	3.16	1.41	0.00	2.00
p4	5.10	3.16	2.00	0.00

L^∞	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Cosine Similarity



For two vectors $\boldsymbol{A}$ and $\boldsymbol{B}$, the cosine similarity is defined as

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

Example:

$\boldsymbol{A} = [3\ 2\ 0\ 5\ 0\ 0\ 0\ 2\ 0\ 0]$
$\boldsymbol{B} = [1\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 2]$

$\boldsymbol{A} \cdot \boldsymbol{B} = 3*1 + 2*0 + 0*0 + 5*0 + 0*0 + 0*0 + 0*0 + 2*1 + 0*0 + 0*2 = 5$
$\|\boldsymbol{A}\| = (3*3 + 2*2 + 0*0 + 5*5 + 0*0 + 0*0 + 0*0 + 2*2 + 0*0 + 0*0)0.5 = 6.481$
$\|\boldsymbol{B}\| = (1*1 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 1*1 + 0*0 + 2*2)\ 0.5 = 2.245$

$s_{cosine} = .3150$

Information retrieval: Cosine similarity is often used for **word count vectors** to compare documents. It compares the distribution of words.

Similarity Between Binary Vectors

- A common situation is that objects, p and q, have only binary attributes
- Compute similarities using the following quantities

  $f01$ = the number of attributes where p was 0 and q was 1

  $f10$ = the number of attributes where p was 1 and q was 0

  $f00$ = the number of attributes where p was 0 and q was 0

  $f11$ = the number of attributes where p was 1 and q was 1

- **Simple Matching** and **Jaccard Coefficients**

  s_{SMC} = number of matches / number of attributes

  $= (f11 + f00) / (f01 + f10 + f11 + f00)$

  s_J = number of 11 matches / number of not-both-zero attribute values

  $= (f11) / (f01 + f10 + f11)$

Note: Jaccard ignores 0s!

SMC versus Jaccard: Example

$$p = [1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0]$$
$$q = [0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 1]$$

$f01 = 2$ (the number of attributes where p was 0 and q was 1)
$f10 = 1$ (the number of attributes where p was 1 and q was 0)
$f00 = 7$ (the number of attributes where p was 0 and q was 0)
$f11 = 0$ (the number of attributes where p was 1 and q was 1)

$$s_{SMC} = \frac{f11 + f00}{f01 + f10 + f11 + f00} = (0 + 7) / (2 + 1 + 0 + 7) = 0.7$$

$$s_J = \frac{f11}{f01 + f10 + f11} = 0 / (2 + 1 + 0) = 0$$

Dis(similarities) With Mixed Types

- Sometimes attributes are of many **different types** (nominal, ordinal, ratio, etc.), but an overall similarity is needed.
- Gower's (dis)similarity:
  - —Ignores missing values
  - —Final (dis)similarity is a weighted sum of variable-wise (dis)similarities
- Calculation:

1. For the kth attribute, compute a similarity, s_k, in the range $[0, 1]$.
2. Define an indicator variable, δ_k, for the k_{th} attribute as follows:

$$\delta_k = \begin{cases} 0 & \text{if the } k^{th} \text{ attribute is a binary asymmetric attribute and both objects have a value of 0, or if one of the objects has a missing values for the } k^{th} \text{ attribute} \\ 1 & \text{otherwise} \end{cases}$$

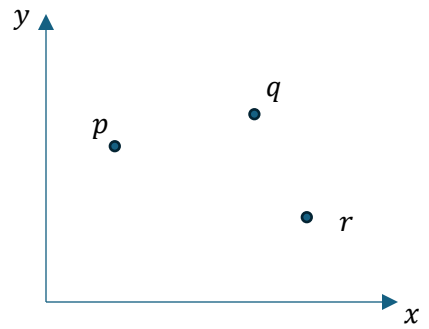
3. Compute the overall similarity between the two objects using the following formula:

$$similarity(p, q) = \frac{\sum_{k=1}^{n} \delta_k s_k}{\sum_{k=1}^{n} \delta_k}$$

Common Properties of a Distance

- Distances, such as the Euclidean distance, have some well-known properties.

1. Positive definiteness:
   - $d(p,q) \geq 0$ for all p, q and
   - $d(p,q) = 0$ only if $p = q$.
2. Symmetry:
   - $d(p,q) = d(q,p)$ for all p, q.
3. Triangle Inequality:
   - $d(p,r) \leq d(p,q) + d(q,r)$ for all p, q, r.



where $d(p,q)$ is the distance (dissimilarity) between points (data objects), p and q.

- A distance that satisfies these properties is a **metric** and forms a **metric space.**

- **Note**: Some **dissimilarities** violate property 3 (Triangular inequality). They are not a metric and some people would not call them a distance.

Common Properties of a Similarity

- Similarities, also have some well-known properties.

$s(p, q) = 1$ (or maximum similarity) only if p = q.

$s(p, q) = s(q, p)$ for all p and q. (Symmetry)

where $s(p, q)$ is the similarity between points (data objects), p and q.

Exercise

	x	y
A	2	1
B	4	3
C	1	1

- Manually calculate the following using the equations in the slides:
  - Euclidean and the Manhattan distances between A and C and A and B
  - Calculate the Cosine similarity between A and C and A and B

- Check your results by using the dist function in R. You will need to Google to find out how to compute the Cosine similarity in R.

Topics

- Attributes/Features
- Types of Data Sets
- Data Quality
- Data Preprocessing
- Similarity and Dissimilarity
- **Density Estimation**



Density

- Density-based clustering requires a notion of density

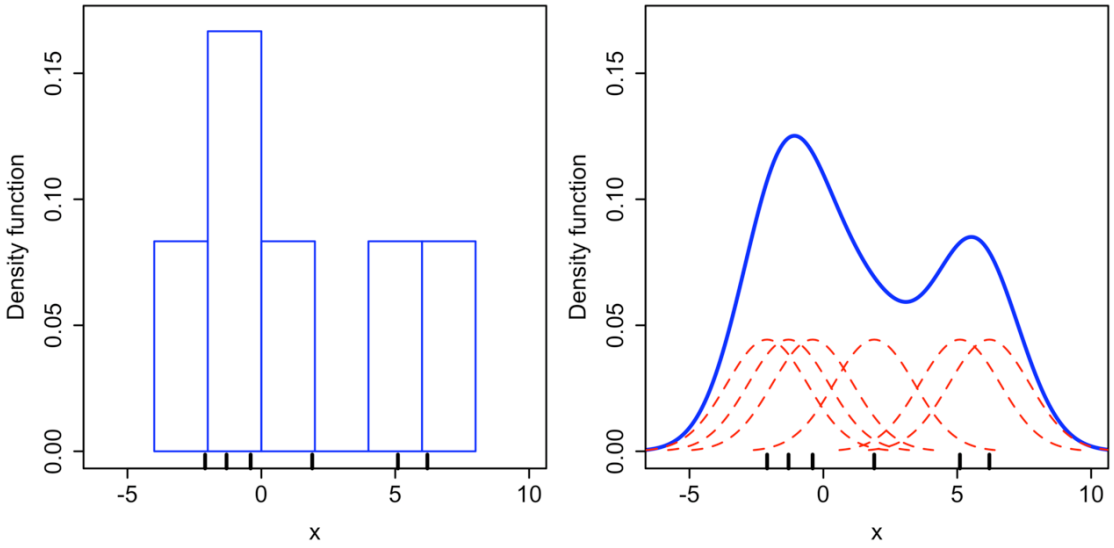
- Examples:
  - Probability density (function) = describes the likelihood of a random variable taking a given value
  - Euclidean density = number of points per unit volume
  - Graph-based density = number of edges compared to a complete graph
  - Density of a matrix = proportion of non-zero entries.

Kernel Density Estimation (KDE)

- KDE is a non-parametric way to estimate the probability density function of a random variable.

$$\widehat{f}_h(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

- K is the kernel (a non-negative function that integrates to one) and $h > 0$ is a smoothing parameter called the bandwidth. Often a Gaussian kernel is used.
- Example:



Euclidean Density – Cell-based

- A Simple approach is to divide region into rectangular cells of equal volume and define density as # of points in each cell.

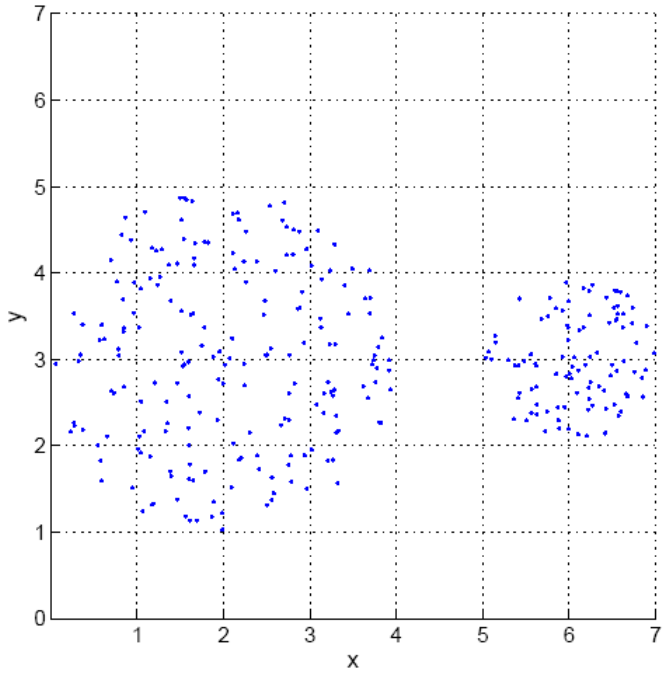


Figure 7.13. Cell-based density.

0	0	0	0	0	0	0
0	0	0	0	0	0	0
4	17	18	6	0	0	0
14	14	13	13	0	18	27
11	18	10	21	0	24	31
3	20	14	4	0	0	0
0	0	0	0	0	0	0

Table 7.6. Point counts for each grid cell.

Euclidean Density – Center-based

- Euclidean density is the number of points within a specified radius around the point. This is also called the **neighborhood** of the point.

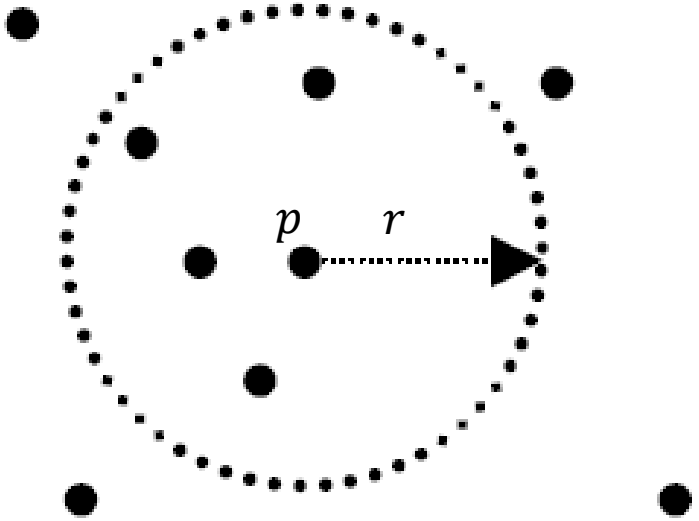


Figure 7.14. Illustration of center-based density.

You should know now about…

- Attributes/Features
- Types of Data Sets
- Data Quality
- Data Preprocessing
- Similarity and Dissimilarity
- Density

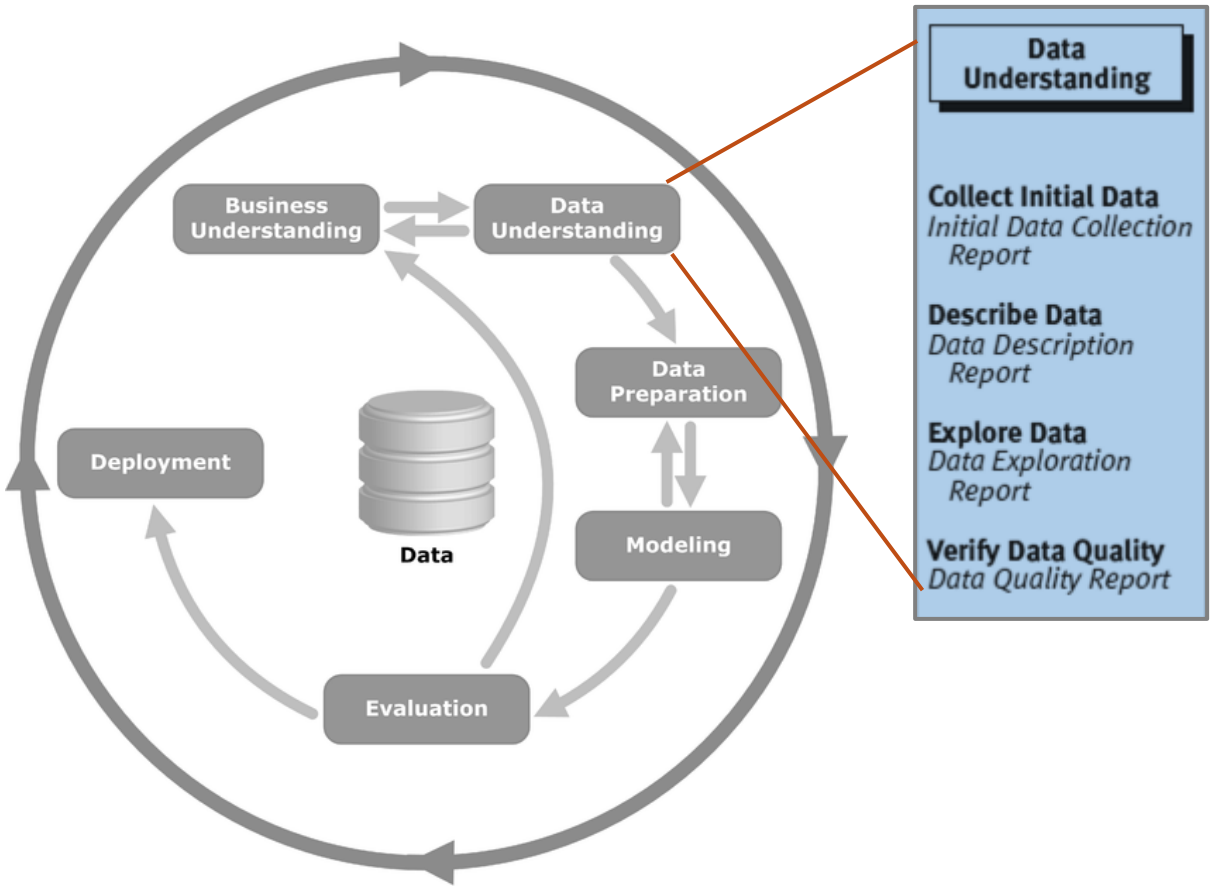


Deep Learning

Exploring Data



Exploring Data in the Data Mining Process



Topics

- **Exploratory Data Analysis**
- Summary Statistics
- Visualization



What is Data Exploration?

“A preliminary exploration of the data to better understand its characteristics.”

- Key motivations of data exploration include
  - —Helping to select the right tool for preprocessing or analysis
  - —Making use of humans’ abilities to recognize patterns.
    People can recognize patterns not captured by data analysis tools

- Related to the area of Exploratory Data Analysis (EDA)
  - —Created by statistician John Tukey
  - —Seminal book is "Exploratory Data Analysis" by Tukey
  - —A nice online introduction can be found in Chapter 1 of the NIST Engineering Statistics Handbook
  - —http://www.itl.nist.gov/div898/handbook/index.htm

Topics

- Exploratory Data Analysis
- **Summary Statistics**
- Visualization



Summary Statistics



Summary statistics are numbers that summarize properties of the data



Summarized properties include location and spread for continuous data

Examples: location - mean
spread - standard deviation



Most summary statistics can be calculated in a single pass through the data

Categorical Features: Frequency and Mode

- The frequency of an attribute value is the percentage of time the value occurs in the data set
  - For example, given the attribute ‘gender’ and a representative population of people, the gender ‘female’ occurs about 60% of the time.

- The mode of an attribute is the most frequent attribute value

Continuous/Ordinal Features:
Measures of Location - Mean and Median

- The mean is the most common measure of the location of a set of points.
- However, the mean is very sensitive to outliers.
- Thus, the median or a trimmed mean is also commonly used.

$$\text{mean}(x) = \overline{x} = \frac{1}{m}\sum_{i=1}^{m} x_i$$

$$\text{median}(x) = \begin{cases} x_{(r+1)} & \text{if } m \text{ is odd, i.e., } m = 2r+1 \\ \frac{1}{2}(x_{(r)} + x_{(r+1)}) & \text{if } m \text{ is even, i.e., } m = 2r \end{cases}$$

Robust against outliers

Measures of Spread: Range and Variance

- Range is the difference between the max and min
- The variance or standard deviation is the most common measure of the spread of a set of points.

$$\text{variance}(x) = s_x^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i - \overline{x})^2$$

- Other measures are often used.

$$\text{AAD}(x) = \frac{1}{m}\sum_{i=1}^{m}|x_i - \overline{x}|$$

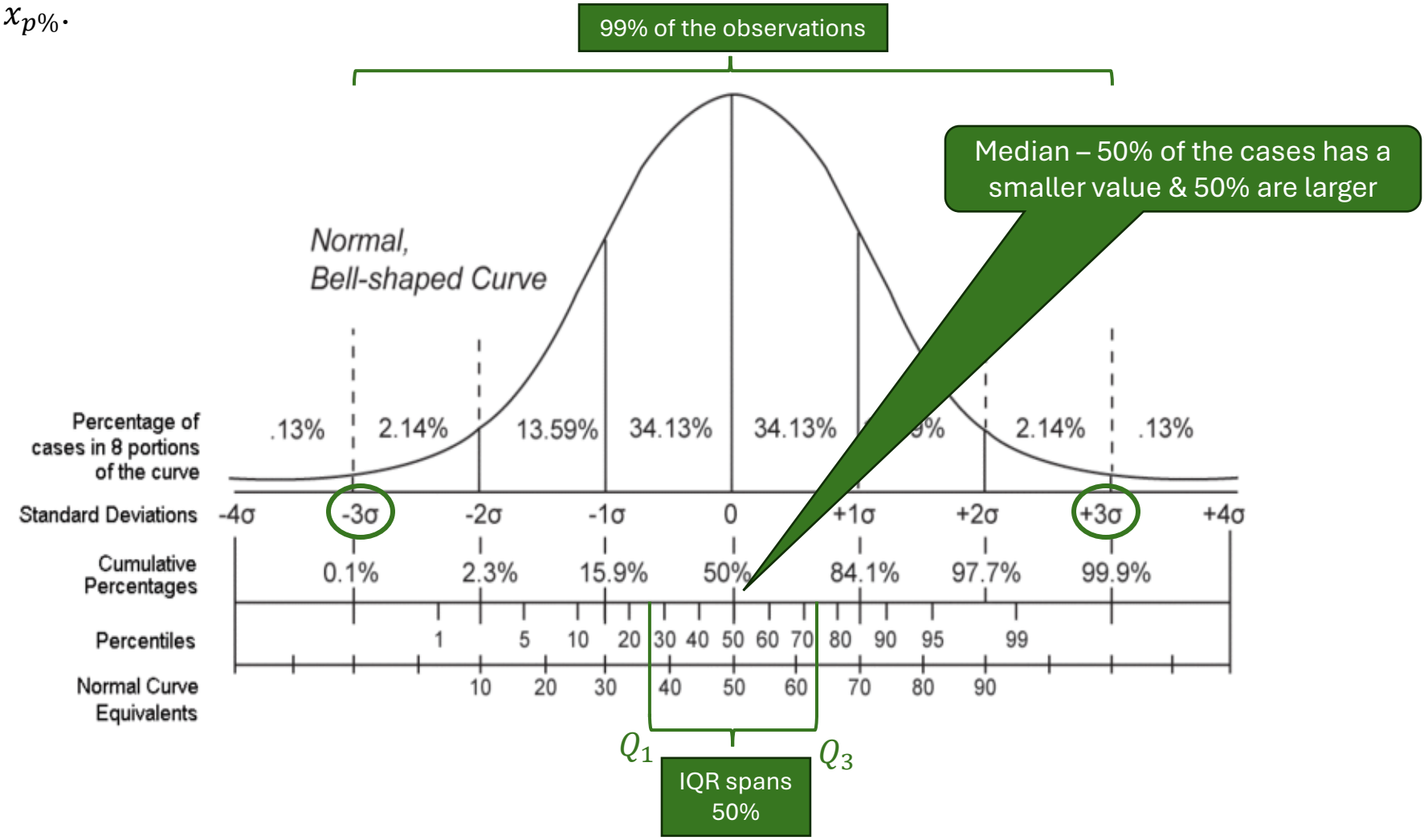
$$\text{MAD}(x) = median\left(\{|x_1 - \overline{x}|, \ldots, |x_m - \overline{x}|\}\right)$$

$$\text{IQR interquartile range}(x) = x_{75\%} - x_{25\%}$$

Robust against outliers

Percentiles of a Distribution

- Given an ordinal or continuous attribute x and a number p between 0 and 100, the pth percentile is a value $x_{p\%}$ of x such that p% of the observed values of x are less than $x_{p\%}$.



Pearson Correlation

- The Pearson correlation coefficient measures the (linear) relationship between two variables.
- To compute Pearson correlation (Pearson's Product Moment Correlation), we standardize data objects, p and q, and then take their dot product

$$\rho = \frac{\mathrm{cov}(X, Y)}{\mathrm{sd}(X)\,\mathrm{sd}(Y)}$$

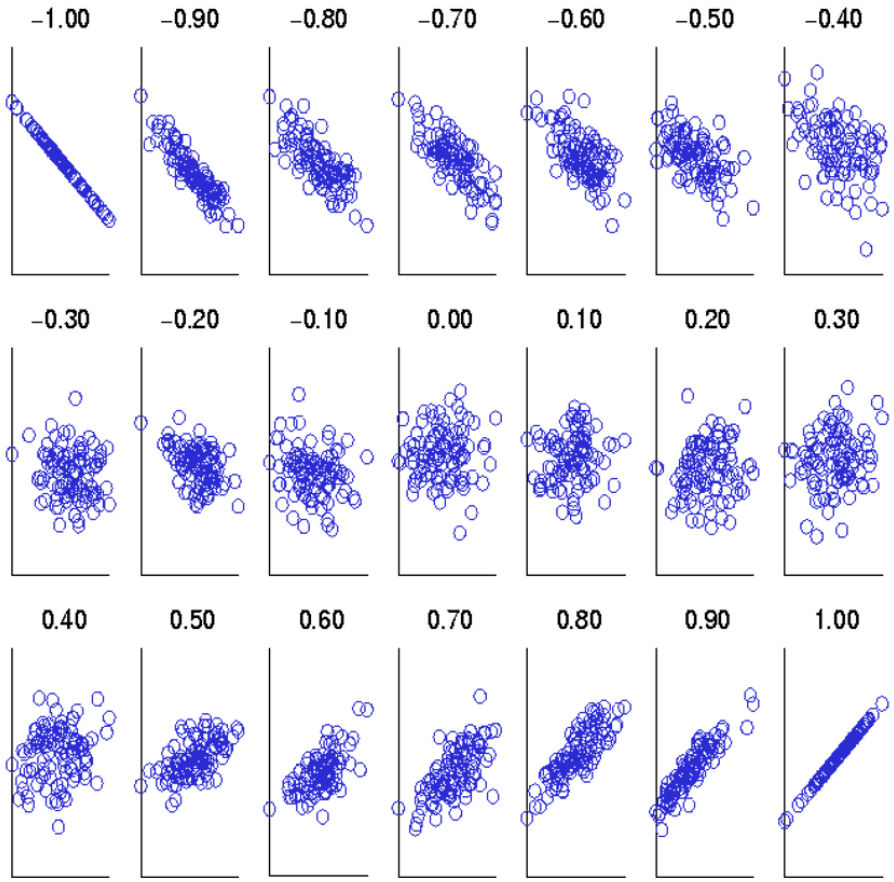
- Estimation:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

- Correlation is often used as a measure of similarity.

Visually Evaluating Correlation

Scatter plots showing data with correlation from –1 to 1.



Rank Correlation

- Measure the degree of similarity between two ratings (e.g., ordinal data).
- Is more robust against outliers and does not assume normality of data or linear relationship like Pearson Correlation.
- Measures (all are between -1 and 1)
  - —Spearman's Rho: Pearson correlation between ranked variables.
  - —Kendall's Tau

$$\tau = \frac{N_s - N_d}{\frac{1}{2}n(n-1)}$$

N_s... concordant pair
N_d... discordant pair

  - —Goodman and Kruskal's Gamma

$$\gamma = \frac{N_s - N_d}{N_s + N_d}$$

Topics

- Exploratory Data Analysis
- Summary Statistics
- **Visualization**



Visualization



Visualization is the conversion of data into **a visual or tabular** format so that the characteristics of the data and the **relationships among data items or attributes** can be analyzed or reported.



Visualization of data is one of the most powerful and appealing techniques for data exploration.

Humans have a well-developed ability to analyze large amounts of information that is presented visually

- Can detect general patterns and trends
- Can detect outliers and unusual patterns

Representation

- Is the mapping of information to a visual format
- Data objects, their attributes, and the relationships among data objects are translated into graphical elements such as **points, lines, shapes, and colors**.

- Examples:
  - Objects are often represented as points.
  - Their attribute values can be represented as the position of the points or the characteristics of the points, e.g., color, size, and shape.
  - If position is used, then the relationships of points, i.e., whether they form groups or a point is an outlier, is easily perceived.

Arrangement

- Is the placement of visual elements within a display
- Can make a large difference in how easy it is to understand the data

Example:

	1	2	3	4	5	6
1	0	1	0	1	1	0
2	1	0	1	0	0	1
3	0	1	0	1	1	0
4	1	0	1	0	0	1
5	0	1	0	1	1	0
6	1	0	1	0	0	1
7	0	1	0	1	1	0
8	1	0	1	0	0	1
9	0	1	0	1	1	0

	6	1	3	2	5	4
4	1	1	1	0	0	0
2	1	1	1	0	0	0
6	1	1	1	0	0	0
8	1	1	1	0	0	0
5	0	0	0	1	1	1
3	0	0	0	1	1	1
9	0	0	0	1	1	1
1	0	0	0	1	1	1
7	0	0	0	1	1	1

The Iris Dataset

Many of the exploratory data techniques are illustrated with the Iris Plant data set.

- Included as a demo datasert in many tools (R, scikit-learn, Rapidminer, ...).
- Can be obtained from the UCI Machine Learning Repository http://www.ics.uci.edu/~mlearn/MLRepository.html
- From the statistician R.A. Fisher
- 150 flowers, three types (classes).
- Four (non-class) attributes



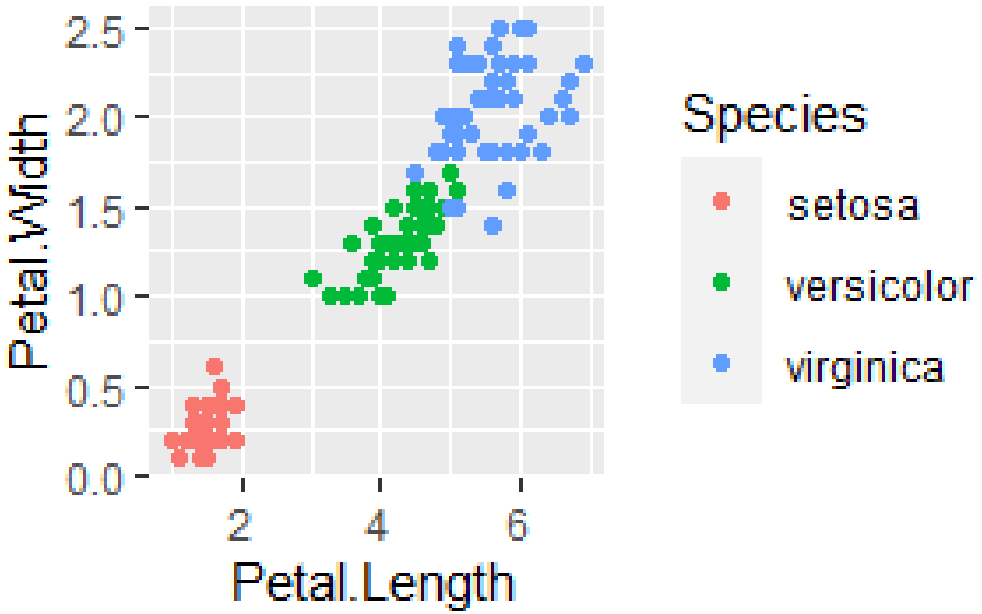




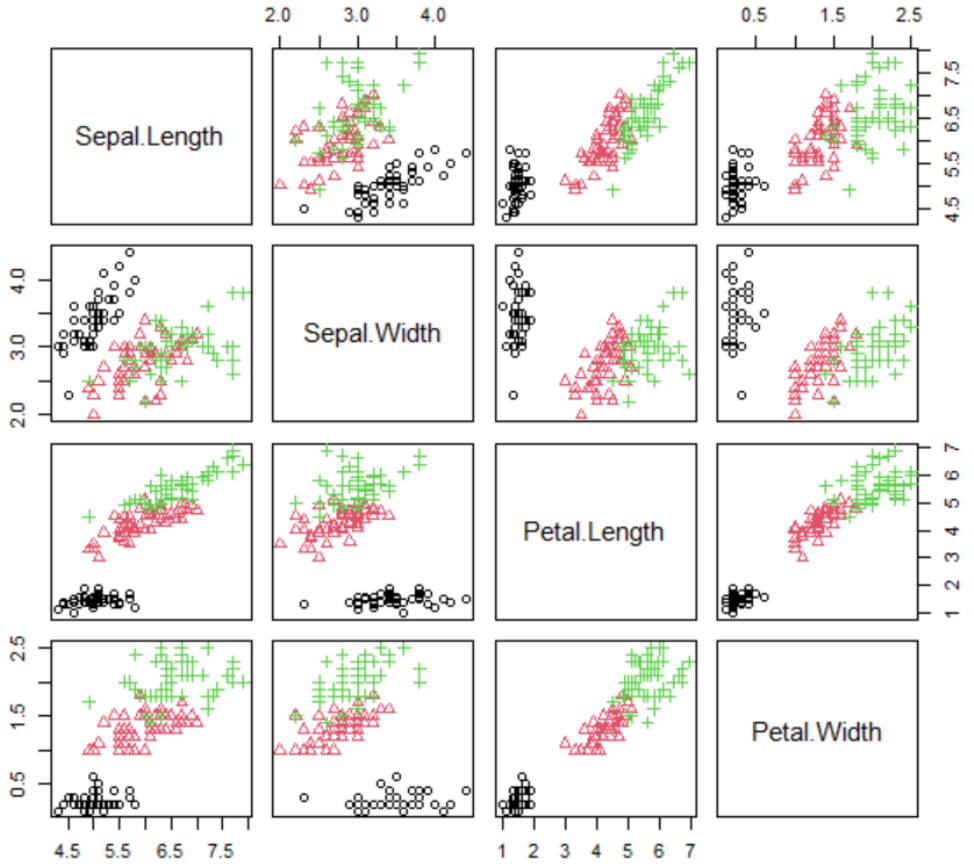
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa

Scatter Plots

- Attributes values determine the position
- Two-dimensional scatter plots most common, but can have three-dimensional scatter plots
- Often additional attributes can be displayed by using the size, shape, and color of the markers that represent the objects



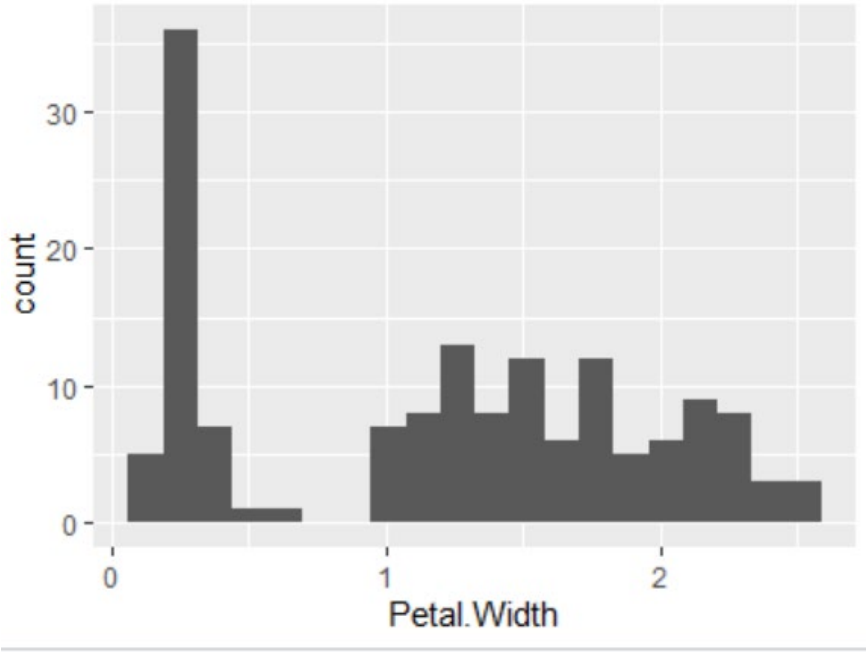
Scatter Plot Array of Iris Attributes

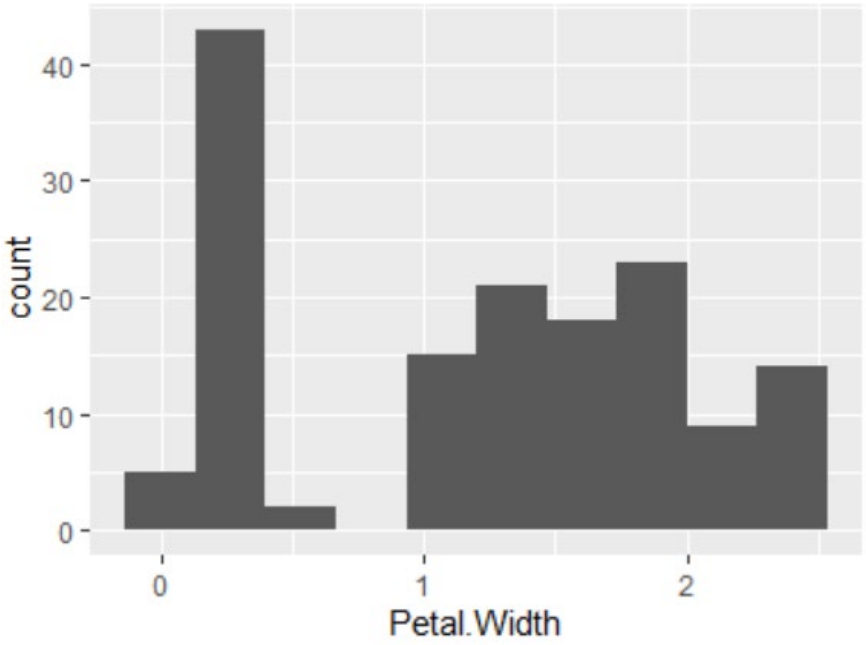


Distribution: Histograms

- Usually shows the distribution of values of a single variable
- Divide the values into bins and show a bar plot of the number of objects in each bin.
- The height of each bar indicates the number of objects
- Shape of histogram depends on the number of bins
- Kernel-based density estimation (KDE) can be used to smooth histograms.

Example: Petal Width (10 and 20 bins, respectively)





Distribution Box Plots

- Invented by J. Tukey as a simplified version of a PDF/histogram that is robust against outliers.
- Used to compare distributions.

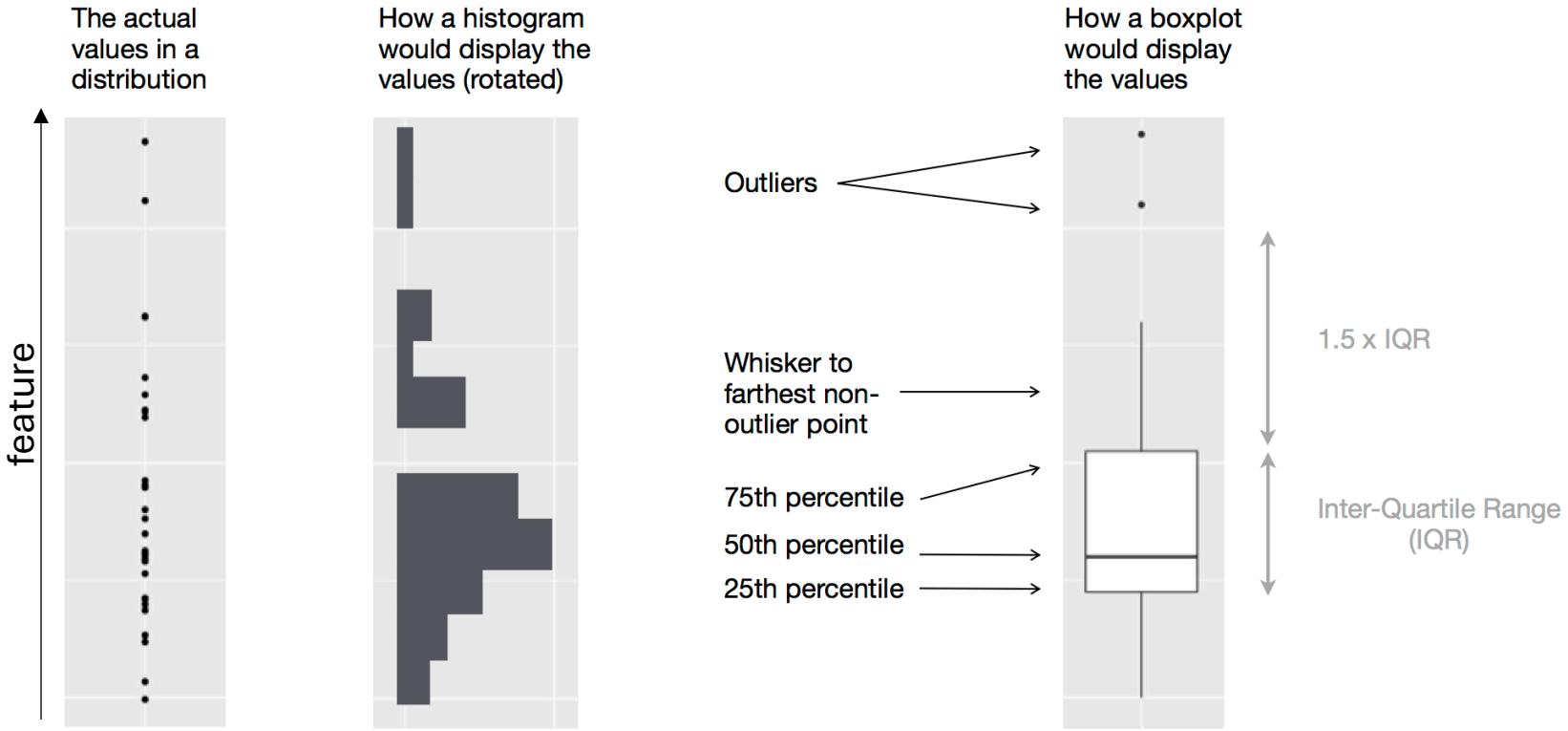
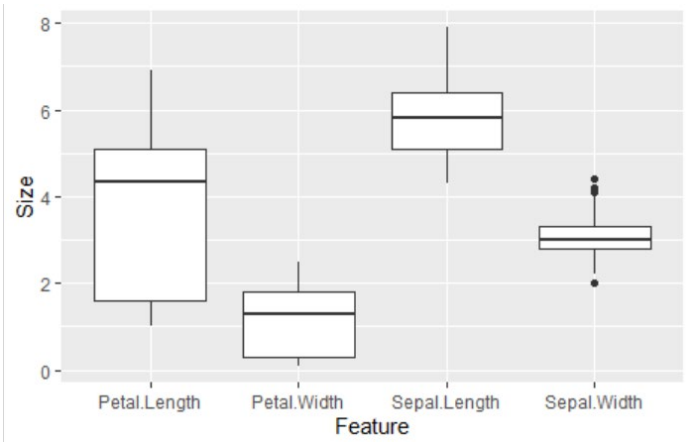


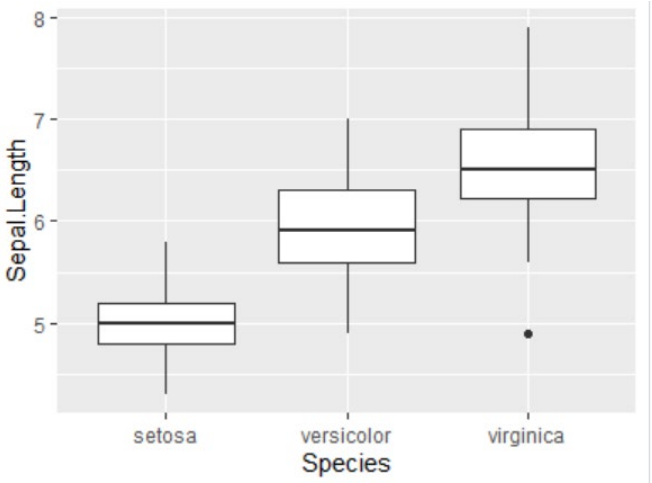
Figure source: Data visualization – R for Data Science (2e)

Examples of Box Plots

- Box plots can be used to compare the distribution of attributes or subgroups.

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa

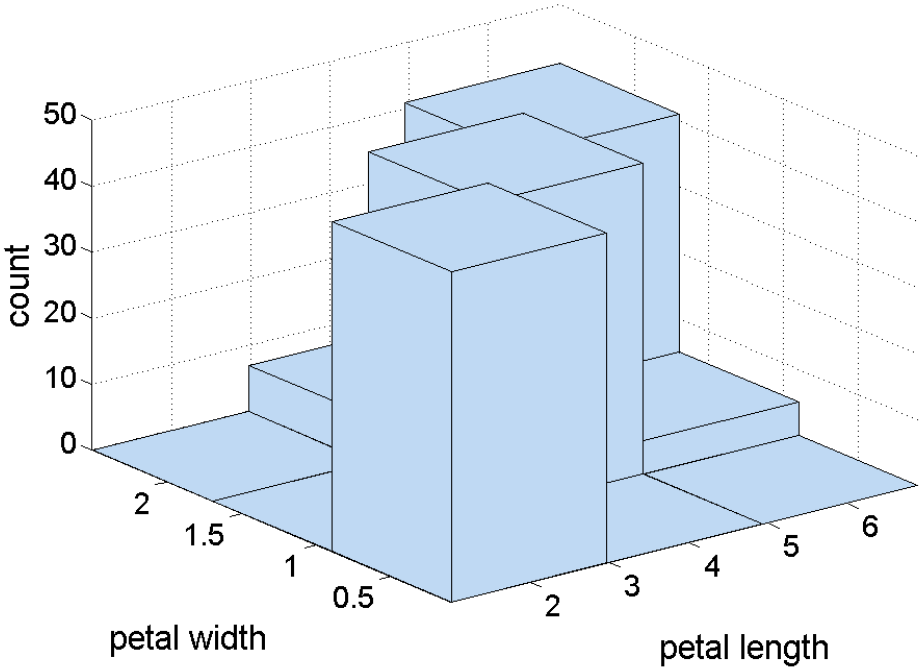




Two-Dimensional Histograms

- Show the joint distribution of the values of two attributes

Example: petal width and petal length. What does this tell us?

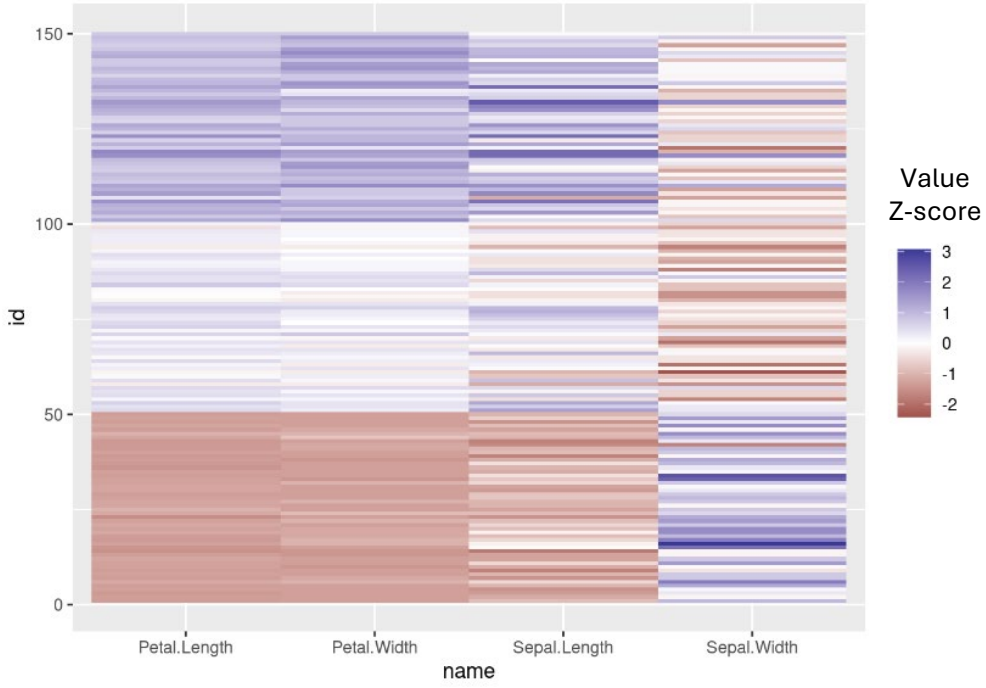


Note: Matrix visualizations are often preferred.

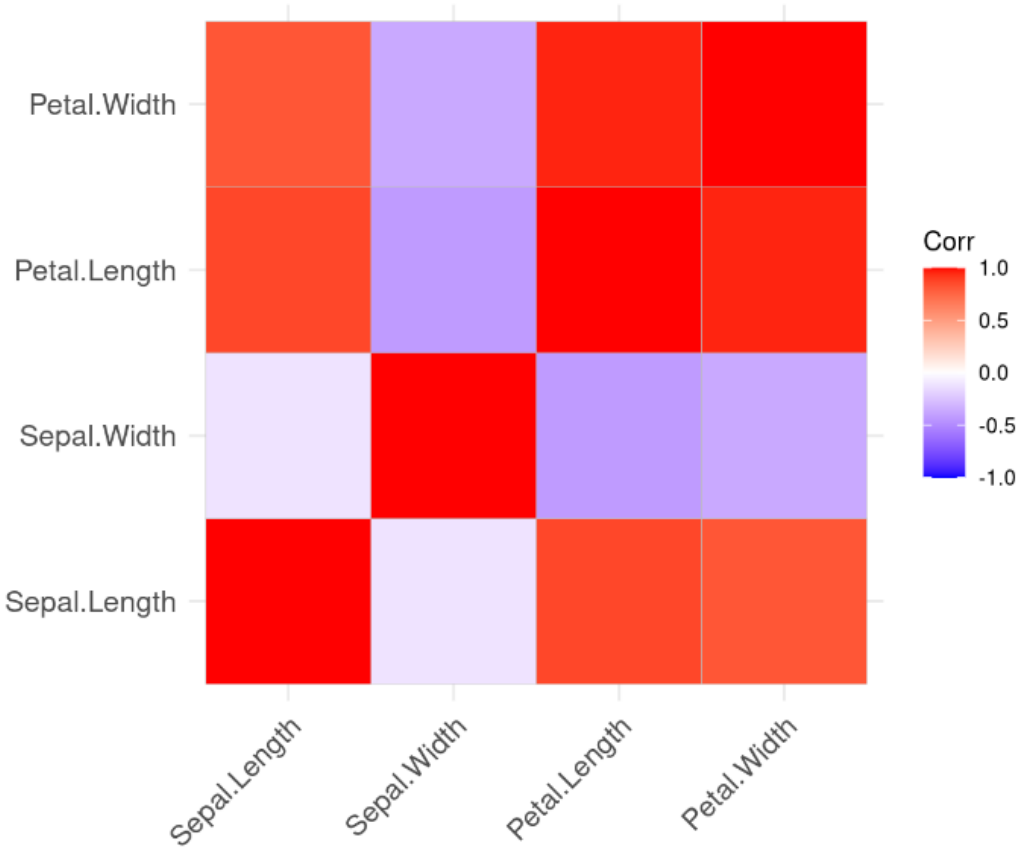
Matrix Plots

- Can plot a data matrix
- Can be useful when objects are sorted according to class
- Typically, the attributes are normalized to prevent one attribute from dominating the plot
- Plots of similarity or distance matrices can also be useful for visualizing the relationships between objects

Example: The Iris Data Matrix



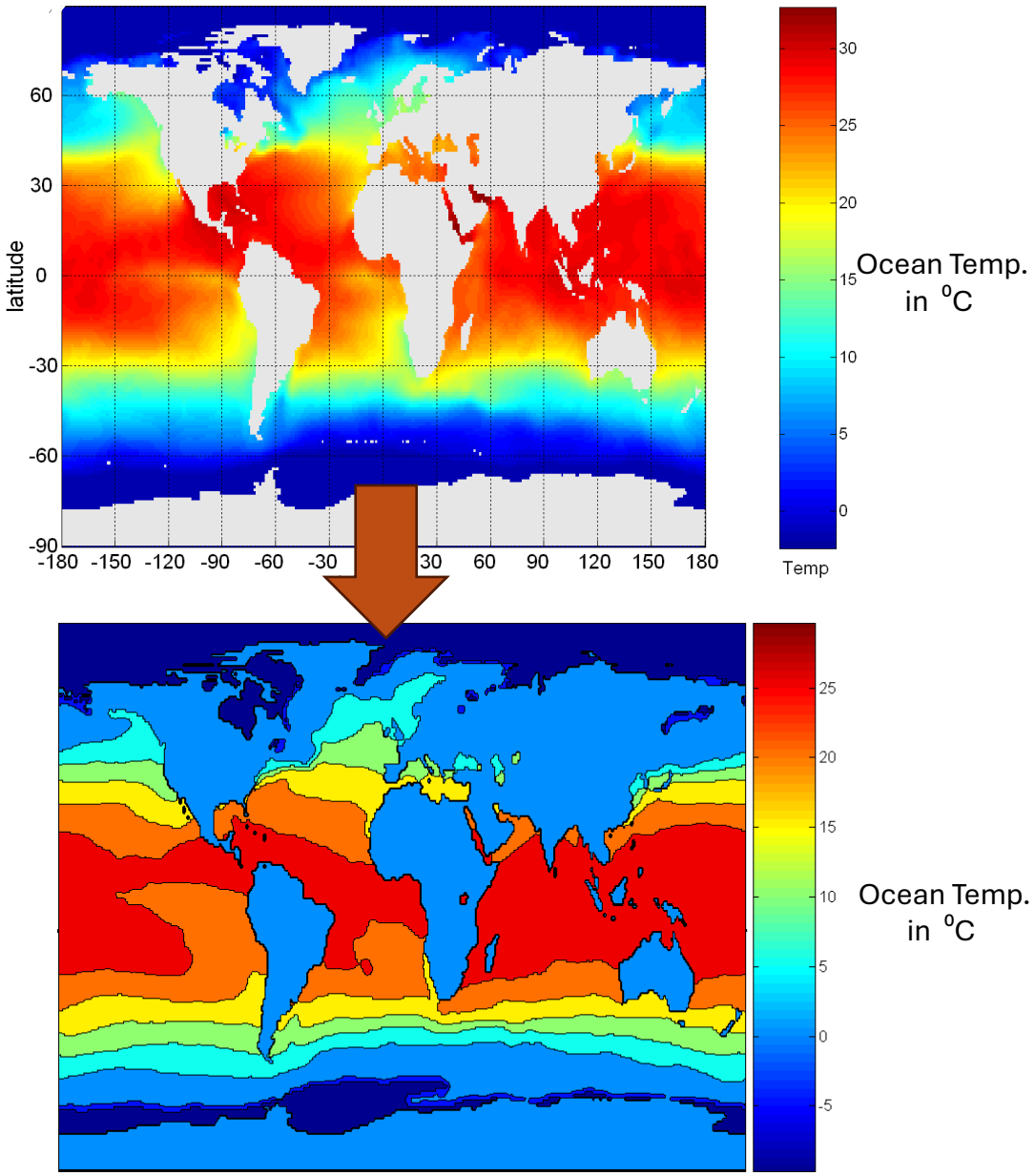
Matrix plot example: The Iris Correlation Matrix



Contour Plots

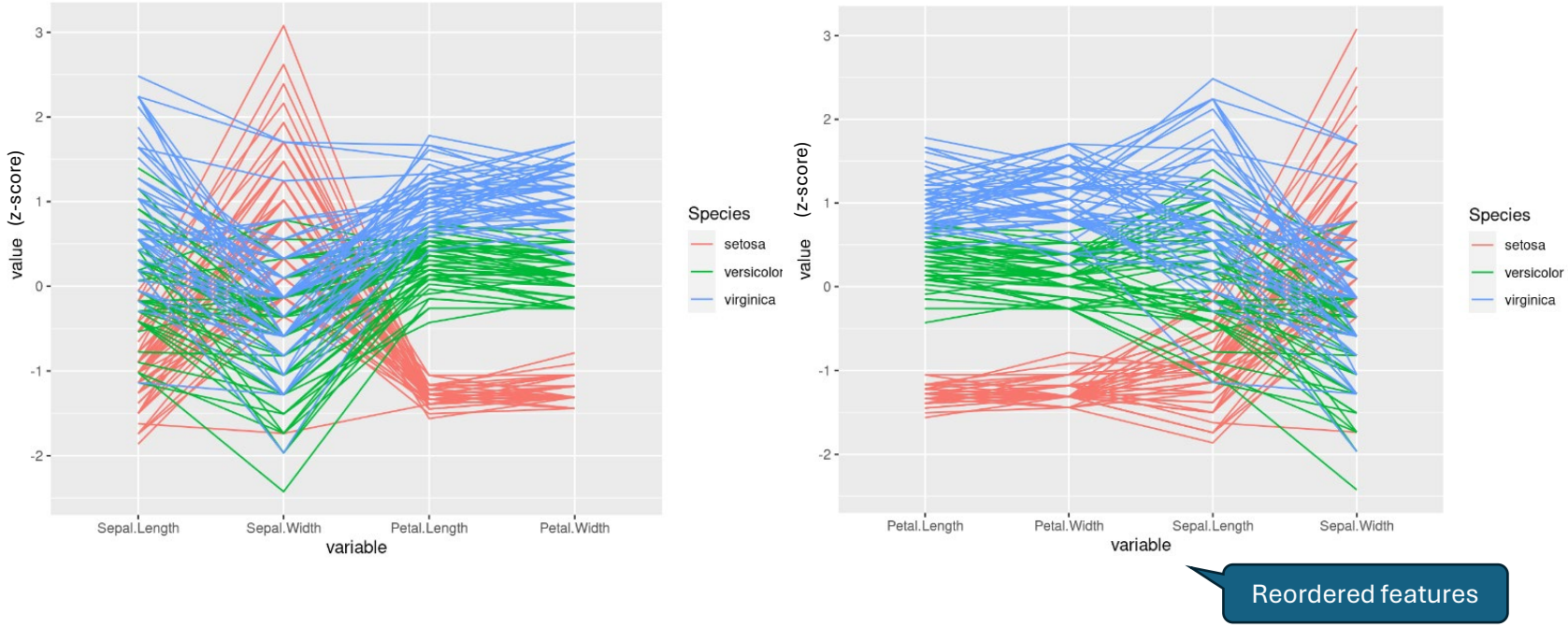
- Useful when a continuous attribute is measured on a **spatial grid**
- They partition the plane into regions of similar values
- The contour lines that form the boundaries of these regions connect points with equal values
- The most common example is contour maps of elevation
- Can also display temperature, rainfall, air pressure, etc.

Sea Surface Temperature (SST) for July 1982



Parallel Coordinates

- Used to plot the attribute values of high-dimensional data
- Instead of using perpendicular axes, use a set of parallel axes
- The attribute values of each object are plotted as a point on each corresponding coordinate axis and the points are connected by a line
- Thus, each object is represented as a line
- Often, the lines representing a distinct class of objects group together, at least for some attributes
- Ordering of attributes is important in seeing such groupings

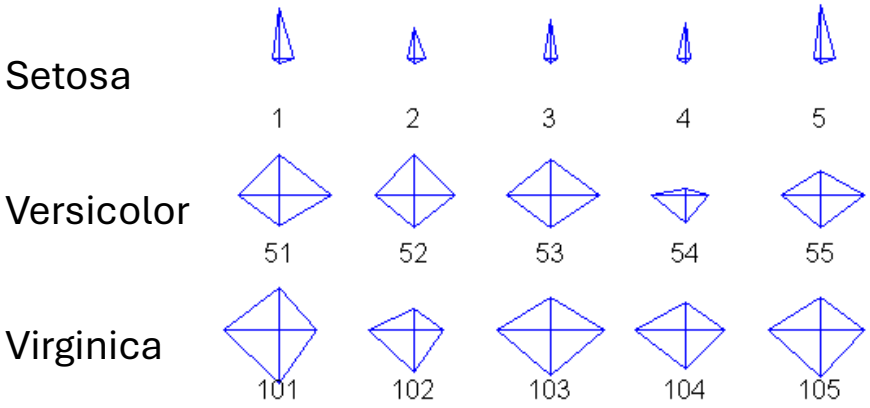


Other Visualization Techniques

- Translate each feature to a feature (a length or size) of a glyph.

Also called radar charts

Star Plots



Chernoff Faces

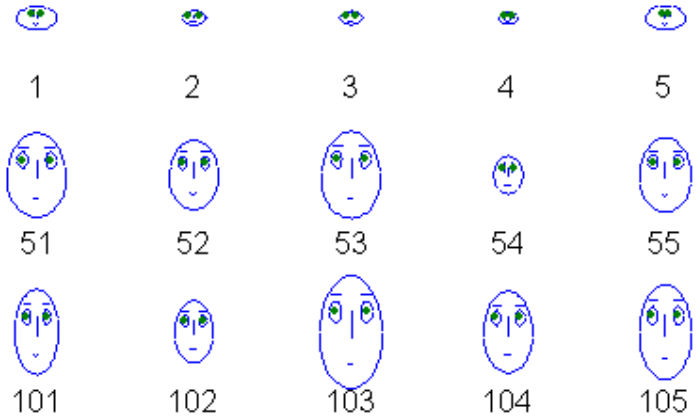
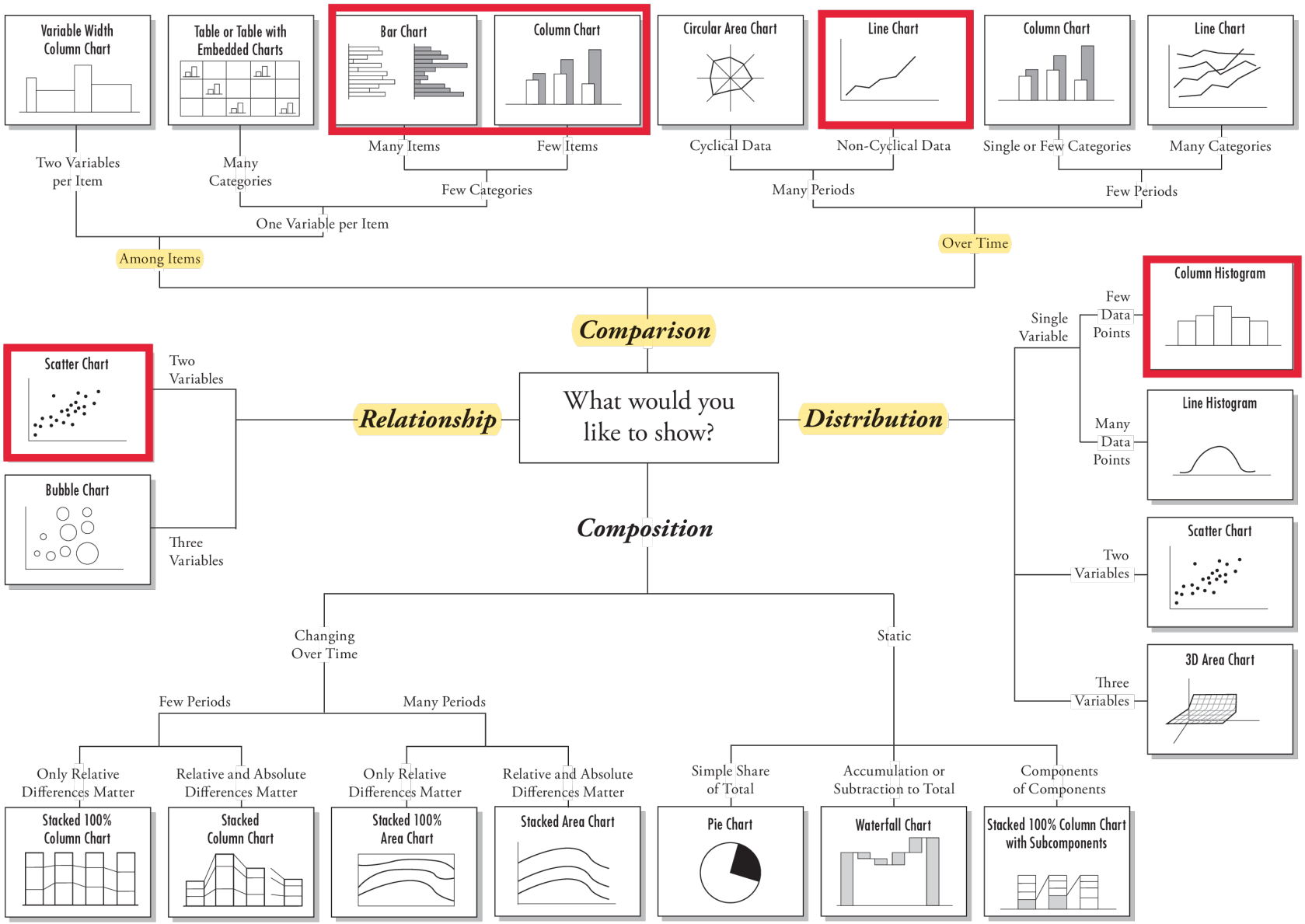


Chart Suggestions—A Thought-Starter



www.ExtremePresentation.com
© 2009 A. Abela — a.v.abela@gmail.com



Conclusion

- Exploring data is the first step when working with data.
- The goal is to:
  1. Understand what data is available.
  2. Assess data distributions and how variables relate to each other.
  3. Assess data quality.

- Understanding the data is necessary to decide on data preparation and modeling.